

Total diagrams: 52

Source folder: docs/02-architecture/mmd-diagrams/architecture

Description folder: docs/02-architecture/diagram-descriptions/mmd-diagrams/architecture

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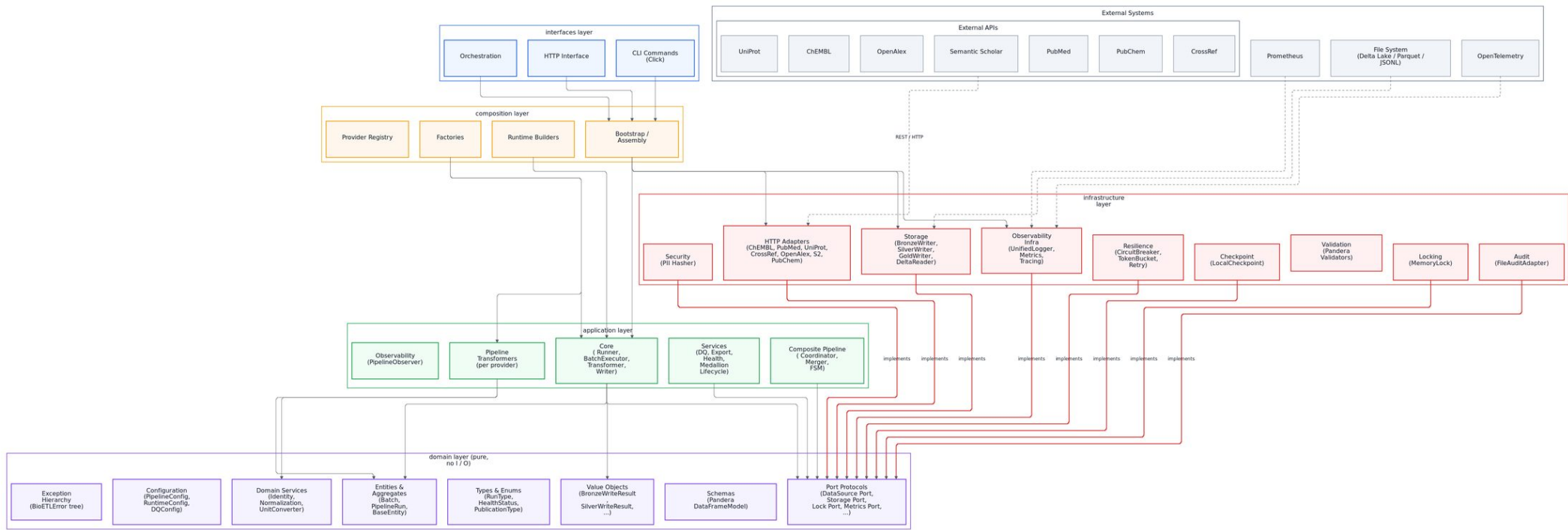
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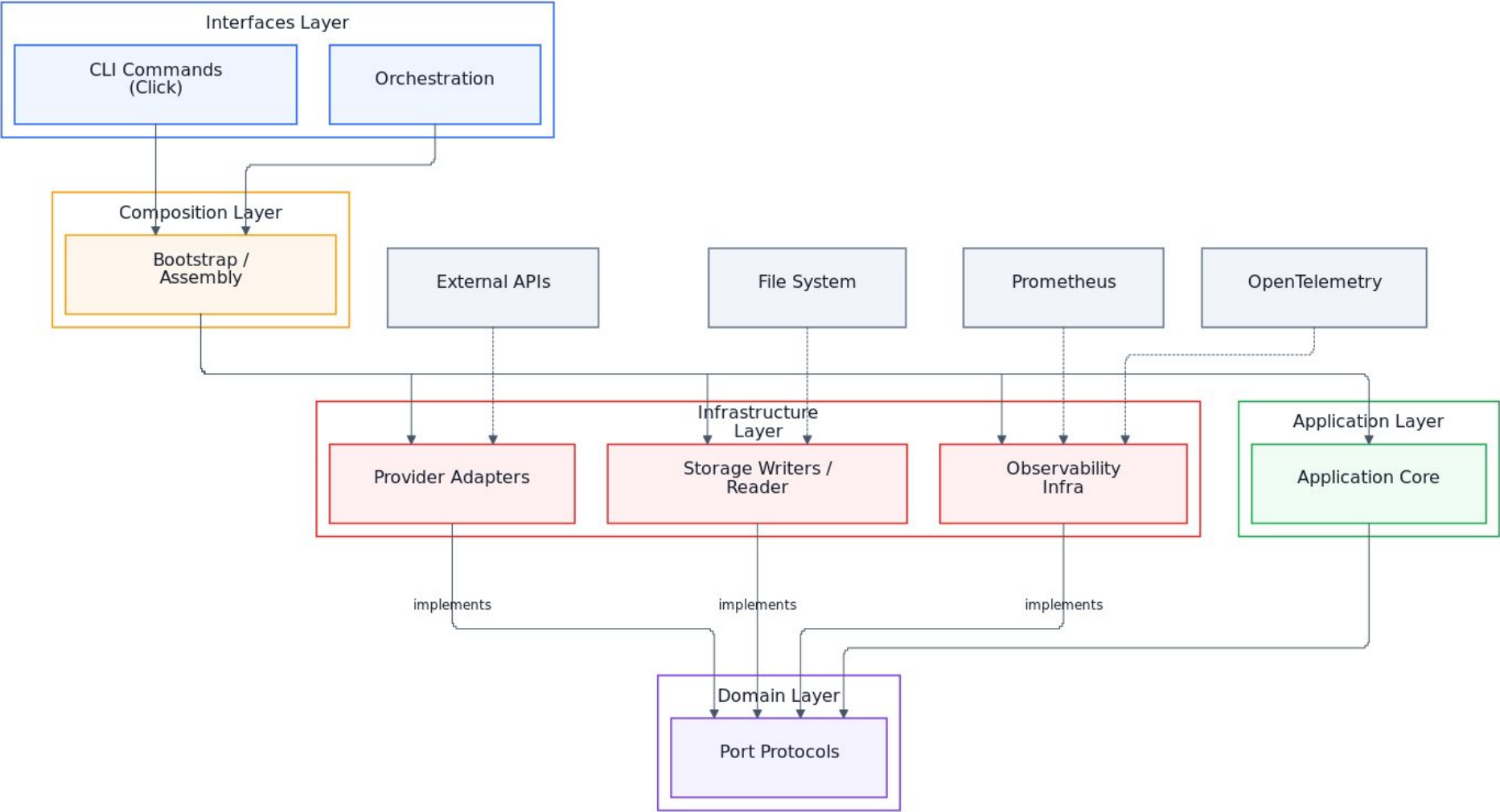
01. High-Level Hexagonal Architecture

ID: 01-high-level-hexagonal Type: flowchart Level: System / Component Date: 2026-02-24



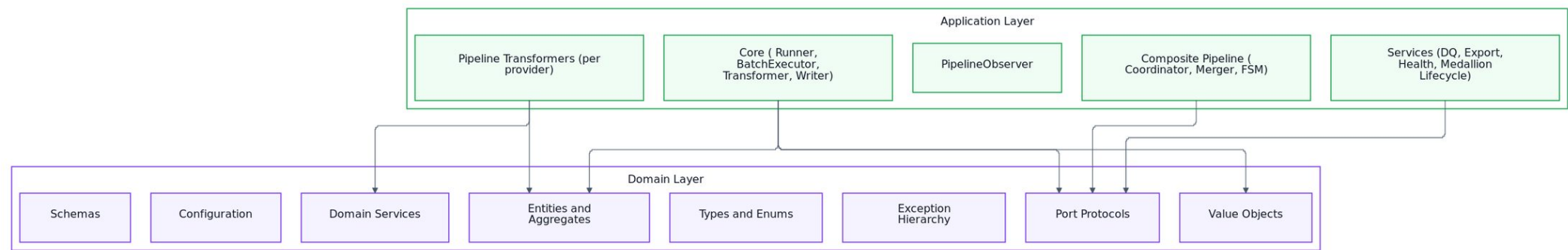
02. Hexagonal Overview

ID: 01a-hexagonal-overview Type: flowchart Level: System / Component Date: 2026-02-25



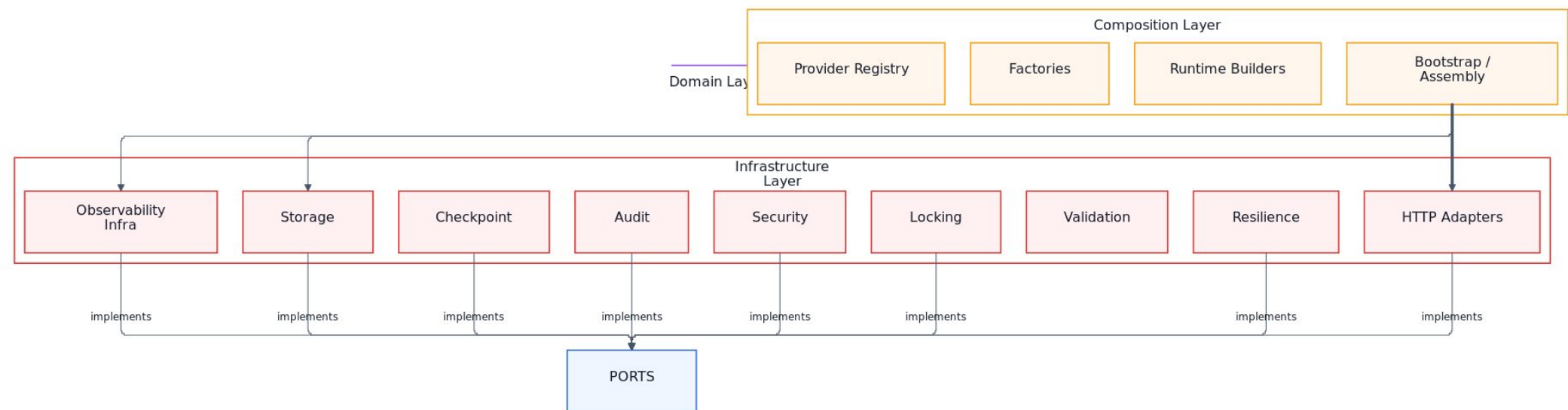
03. Hexagonal Domain and Application

ID: 01b-hexagonal-domain-app Type: flowchart Level: System / Component Date: 2026-02-25



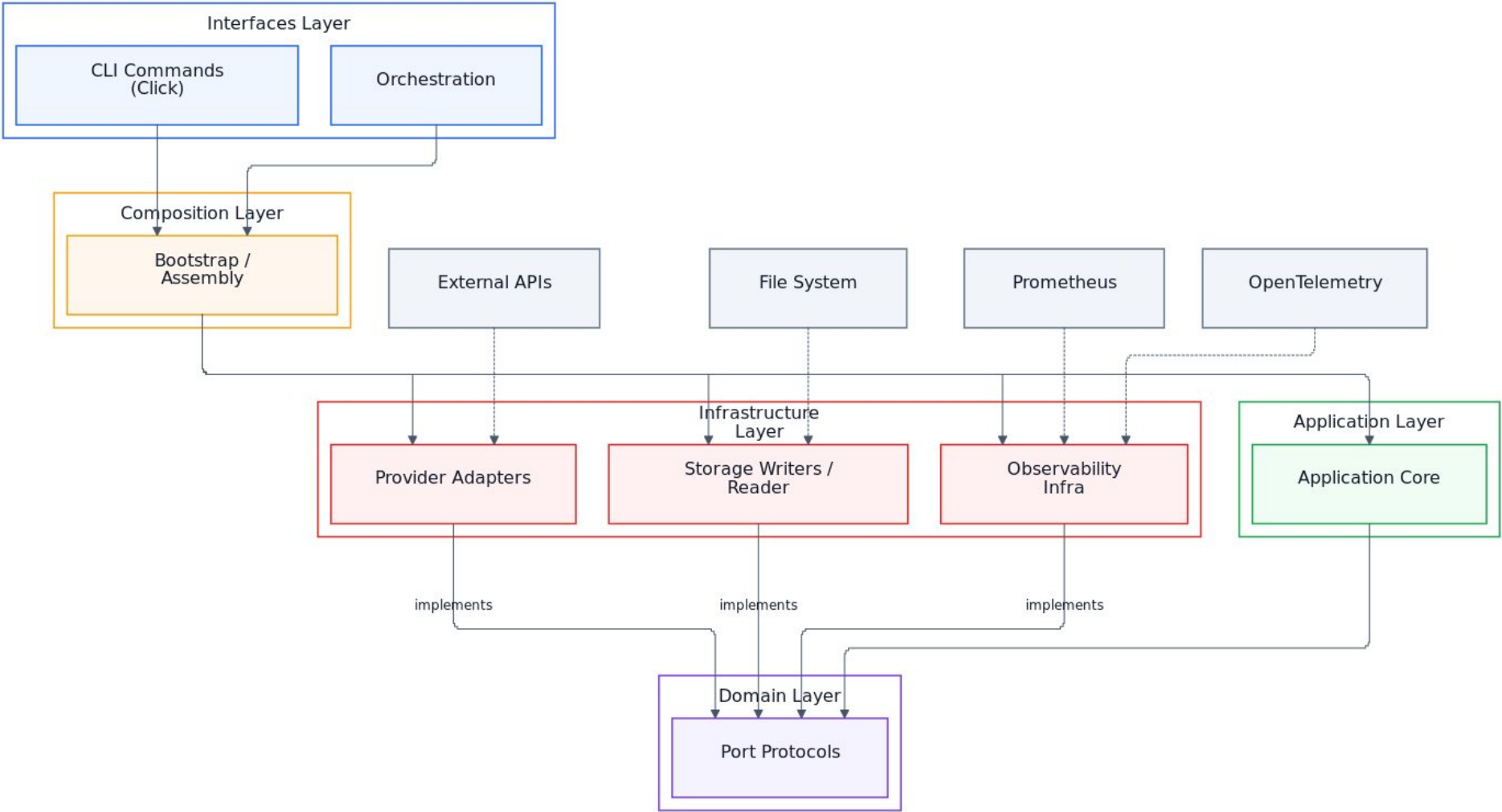
04. Hexagonal Infrastructure and Composition

ID: 01c-hexagonal-infra-comp Type: flowchart Level: System / Component Date: 2026-02-25



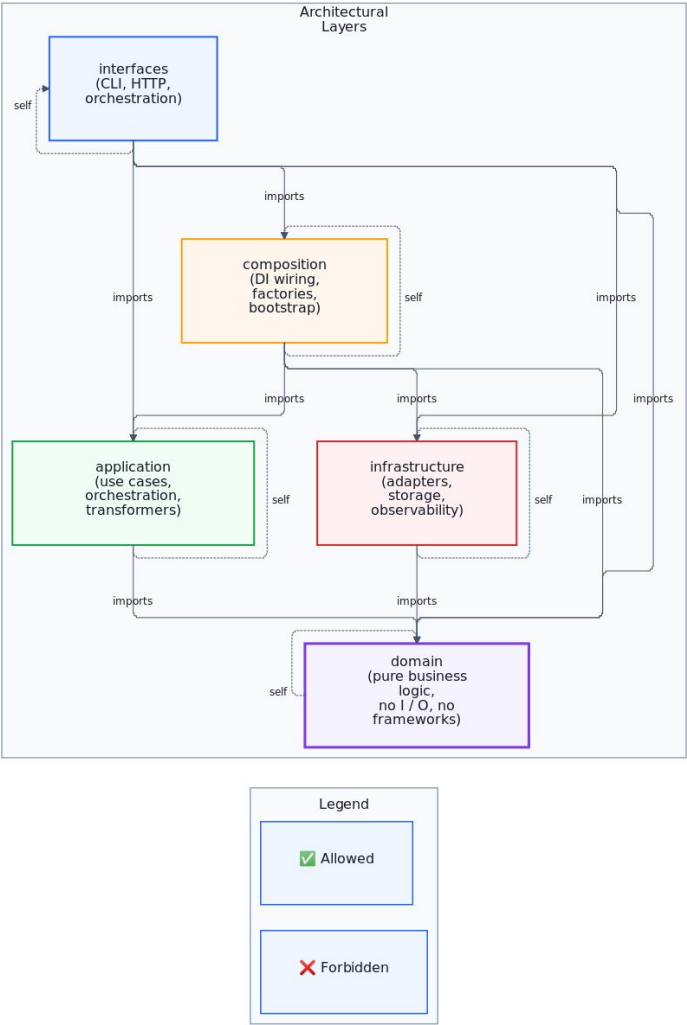
05. Hexagonal Overview (Rounded Nodes)

ID: 01d-hexagonal-overview-rounded Type: flowchart Level: System / Component Date: 2026-03-01



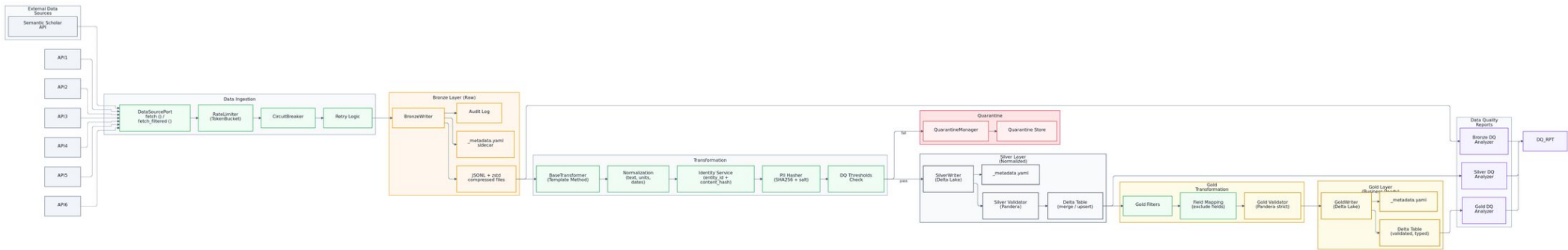
06. Layer Dependency Matrix (ARCH-001)

ID: 02-layer-dependency-matrix Type: flowchart Level: System / Component Date: 2026-02-24



07. Medallion Architecture Data Flow (Bronze → Silver → Gold)

ID: 03-medallion-data-flow Type: flowchart Level: System / Component Date: 2026-02-24



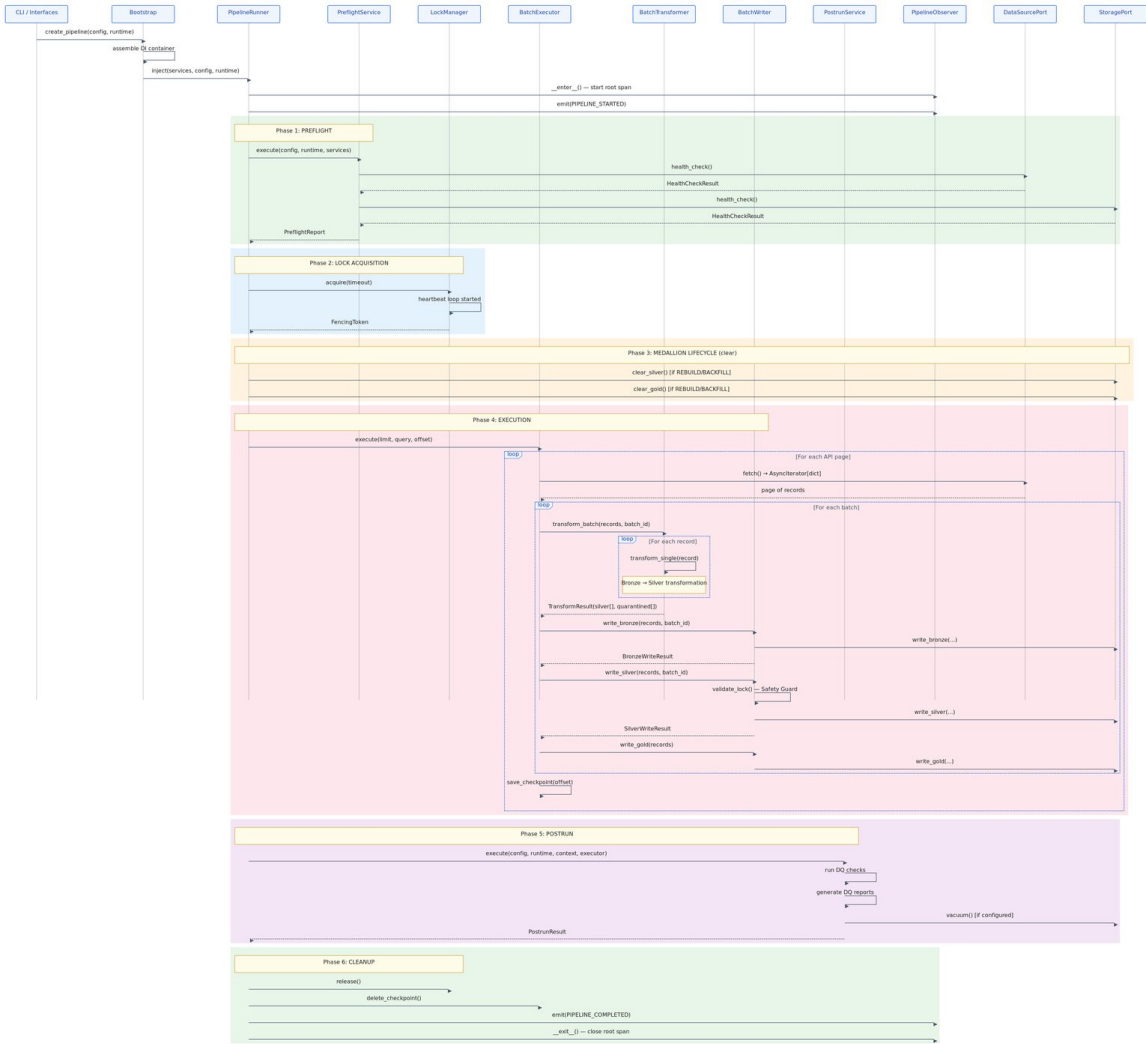
08. Medallion Layers Overview

ID: 03a-medallion-layers-overview Type: flowchart Level: System / Component Date: 2026-02-27



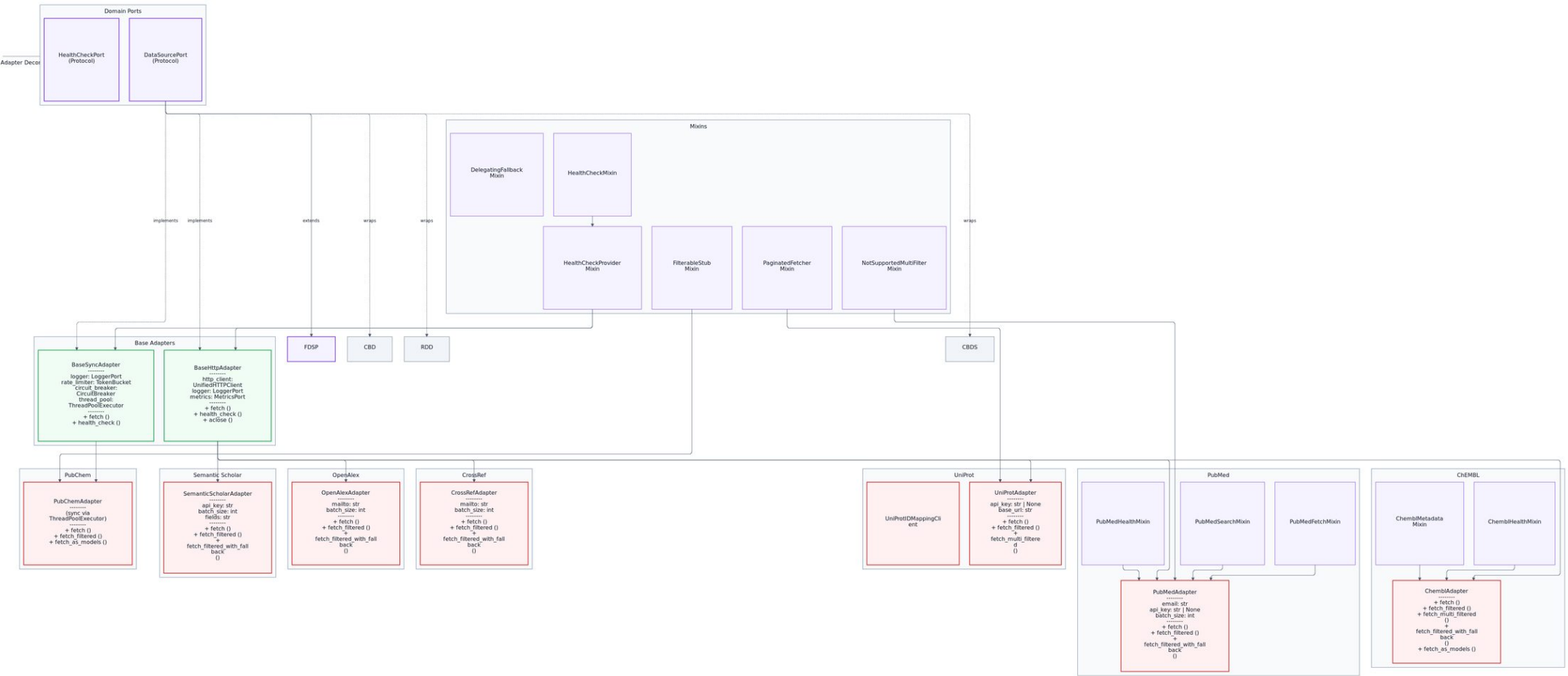
09. Pipeline Execution Lifecycle

ID: 04-pipeline-execution-flow Type: sequenceDiagram Level: System / Component Date: 2026-02-24



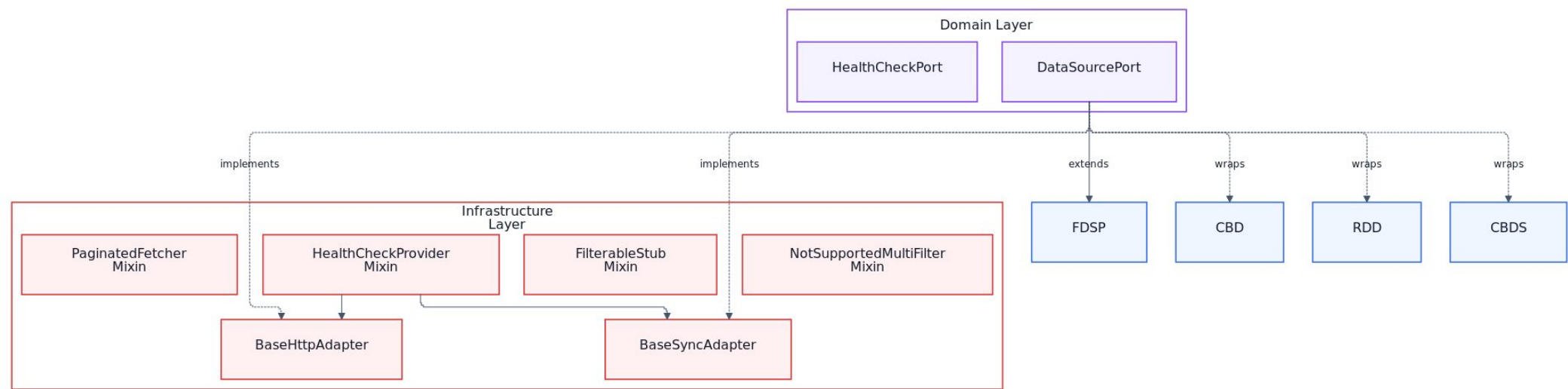
10. Provider Adapter Hierarchy

ID: 05-provider-adapter-hierarchy Type: flowchart Level: System / Component Date: 2026-02-24



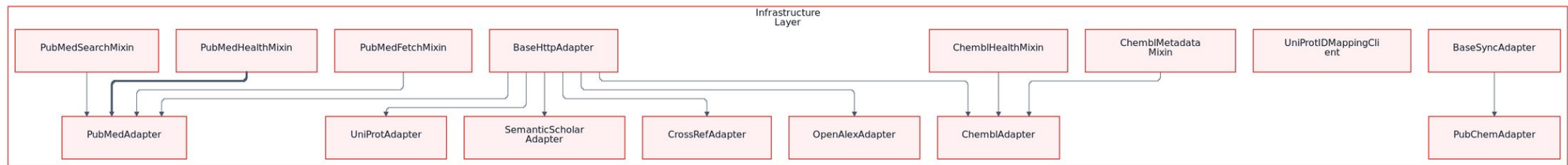
11. Adapter Hierarchy: Base Types

ID: 05a-adapter-hierarchy-base Type: flowchart Level: System / Component Date: 2026-02-25



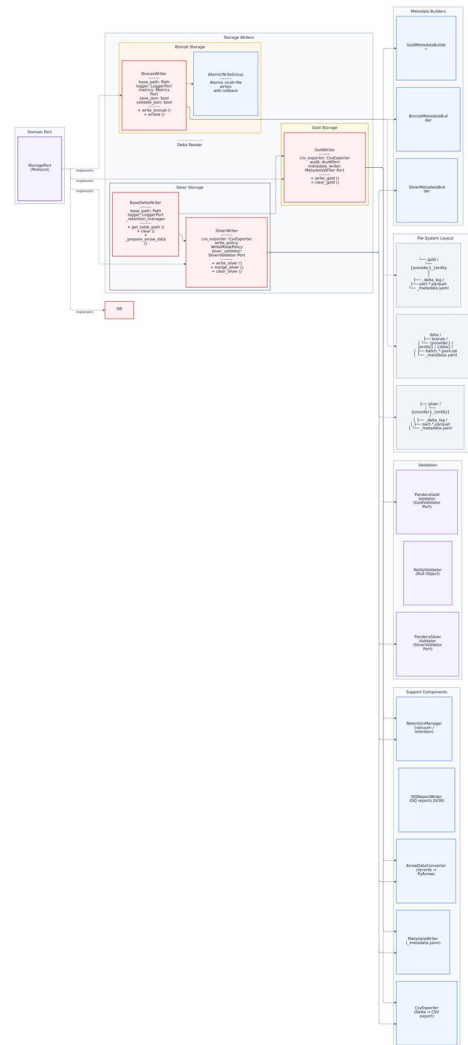
12. Adapter Hierarchy: Provider Implementations

ID: 05b-adapter-hierarchy-providers Type: flowchart Level: System / Component Date: 2026-02-25



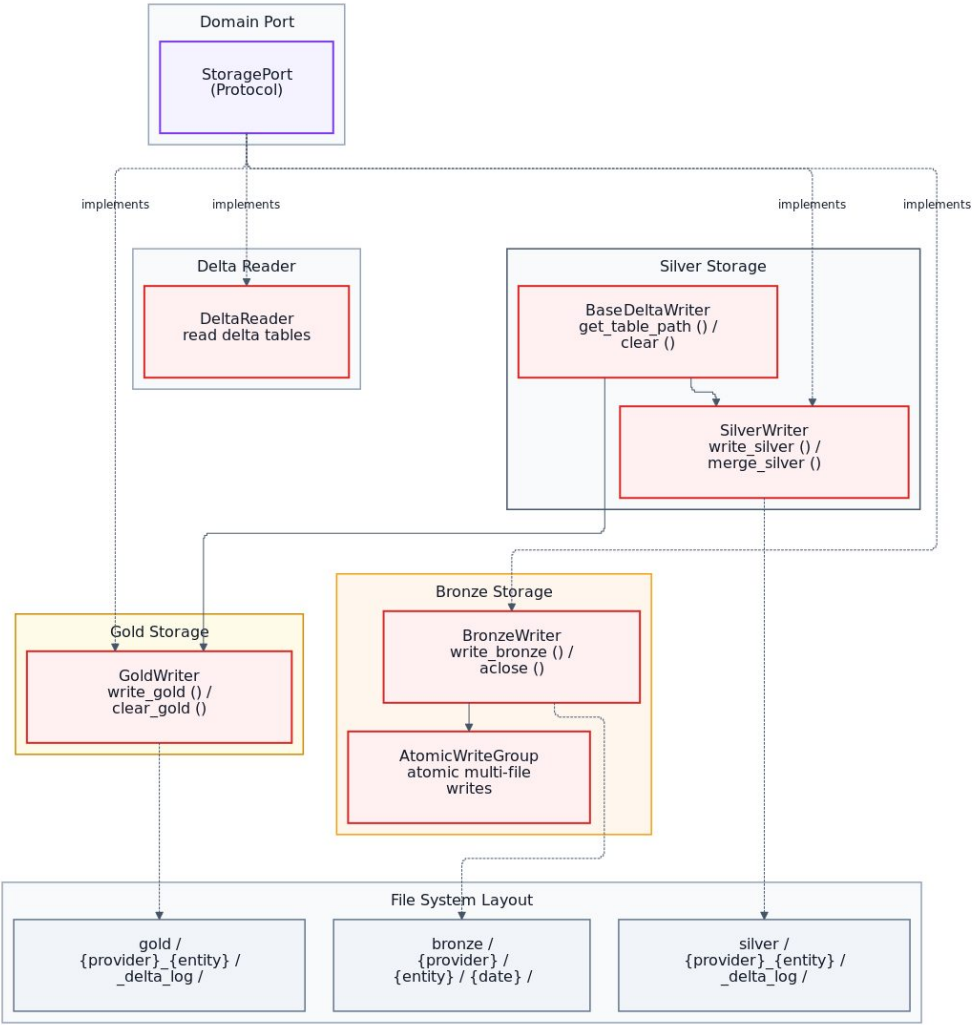
13. Storage Layer Components

ID: 06-storage-layer Type: flowchart Level: System / Component Date: 2026-02-24



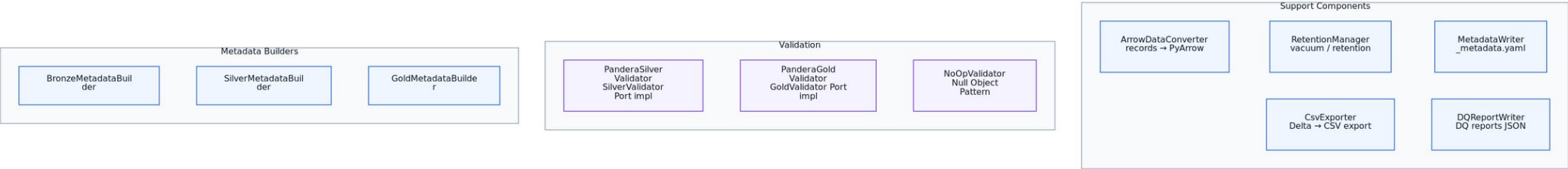
14. Storage Writers

ID: 06a-storage-writers Type: flowchart Level: System / Component Date: 2026-02-27



15. Storage Support Components

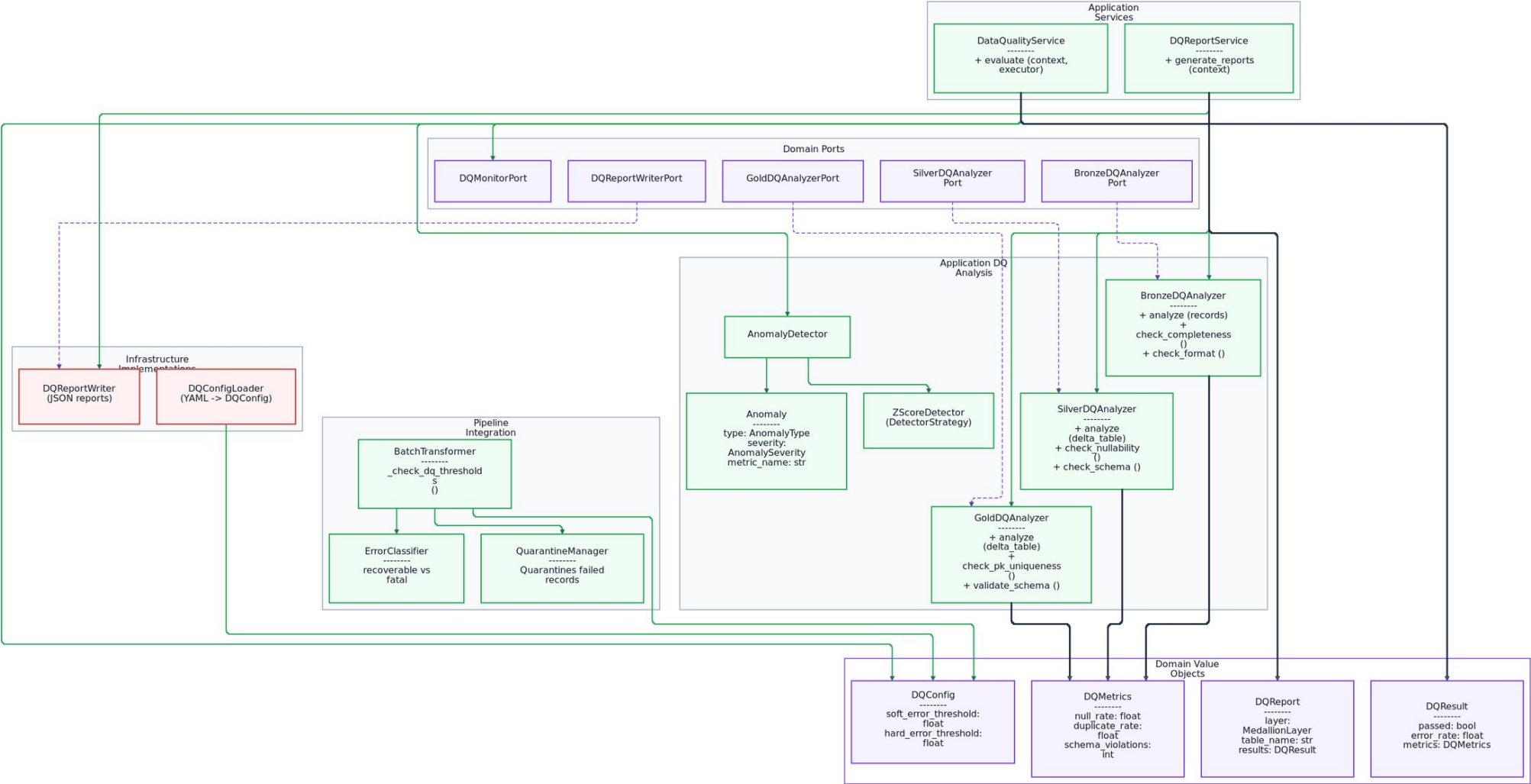
ID: 06b-storage-support Type: flowchart Level: System / Component Date: 2026-02-27



16. Data Quality (DQ) System

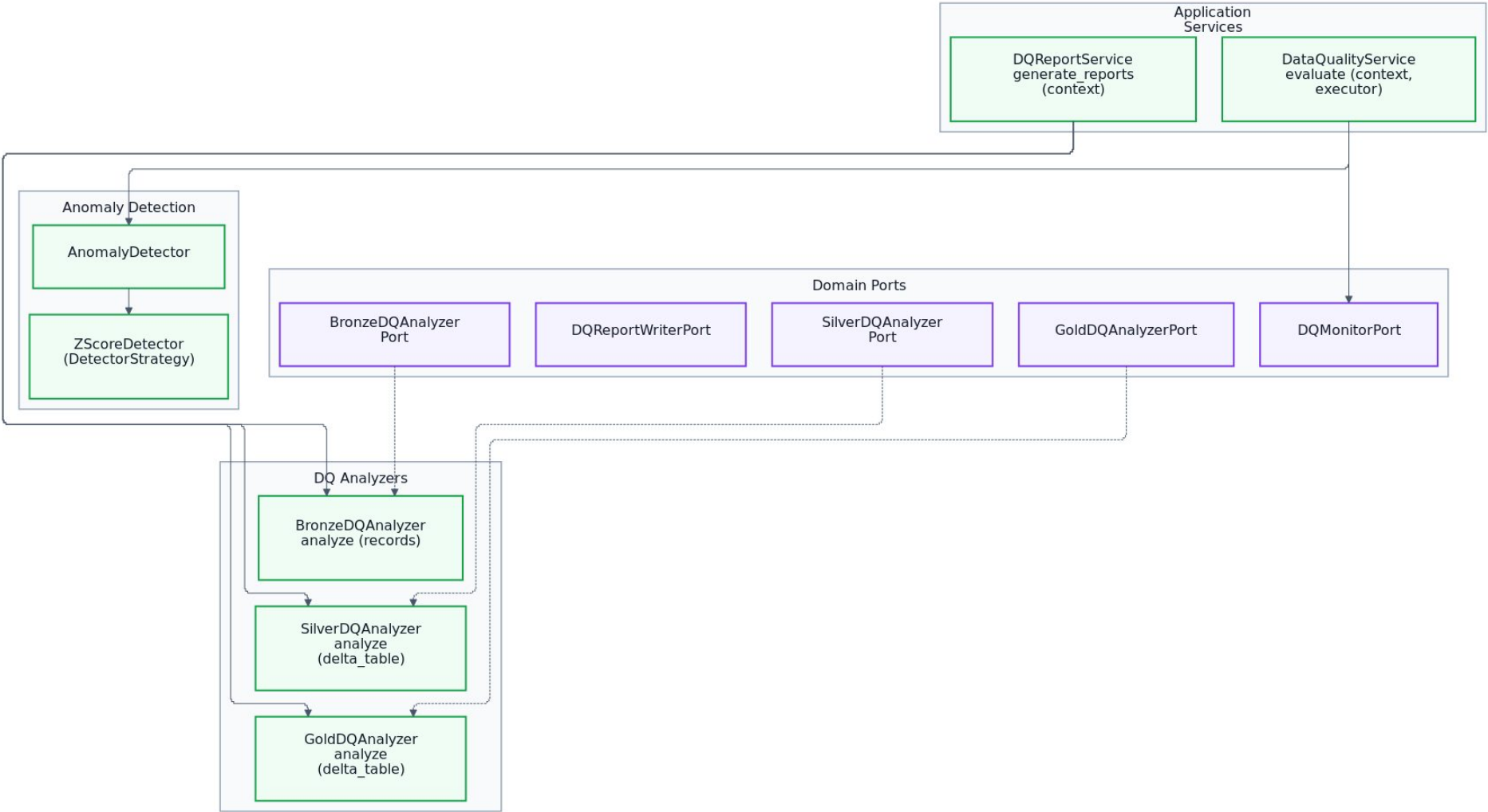
ID: 07-dq-system Type: flowchart Level: System / Component Date: 2026-02-27

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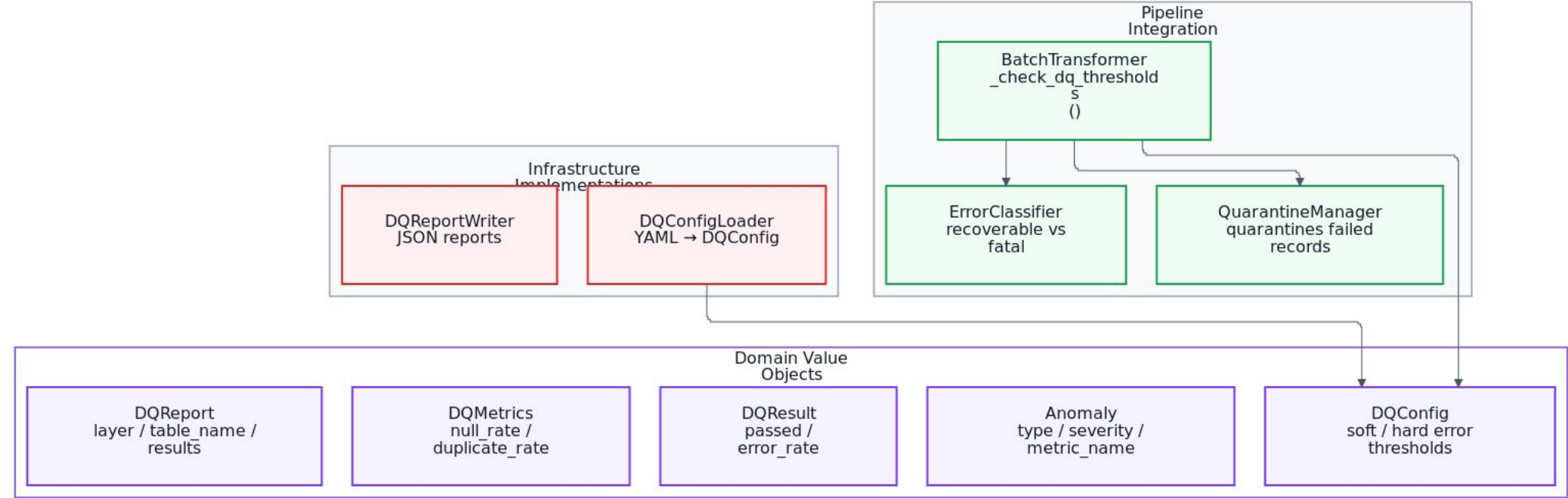
17. DQ Analysis Services

ID: 07a-dq-analysis Type: flowchart Level: System / Component Date: 2026-02-27



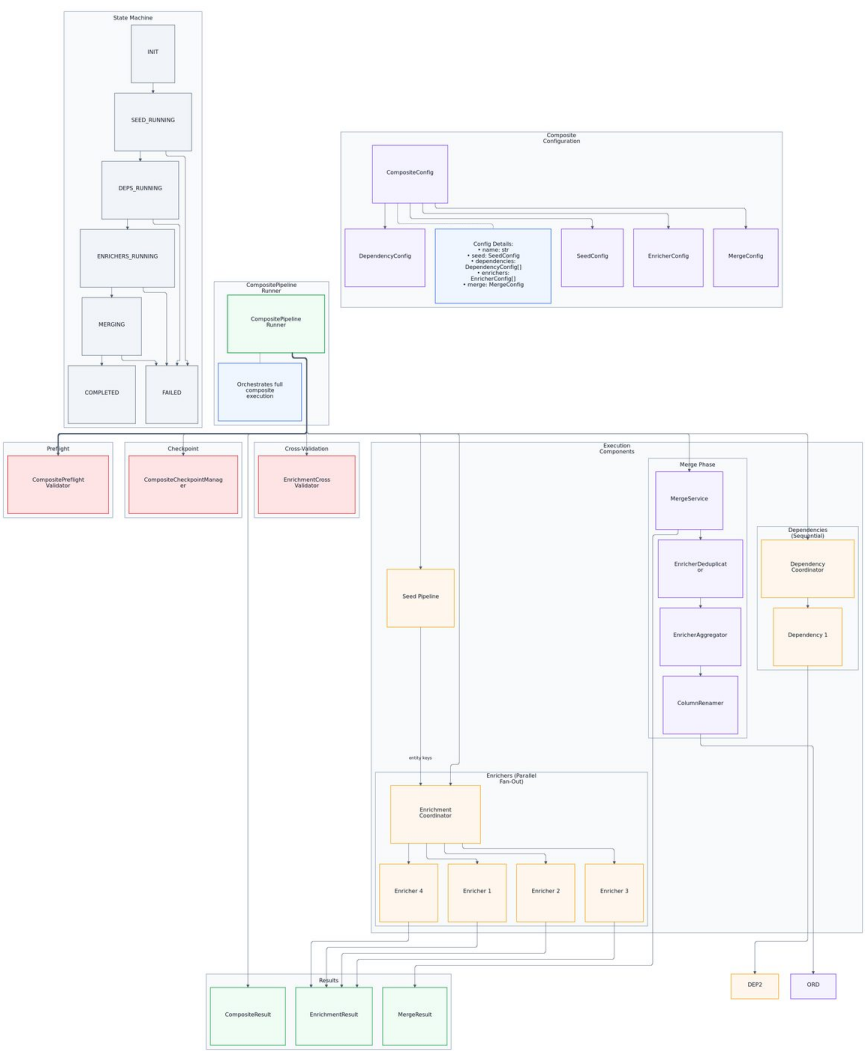
18. DQ Pipeline Integration

ID: 07b-dq-pipeline Type: flowchart Level: System / Component Date: 2026-02-27



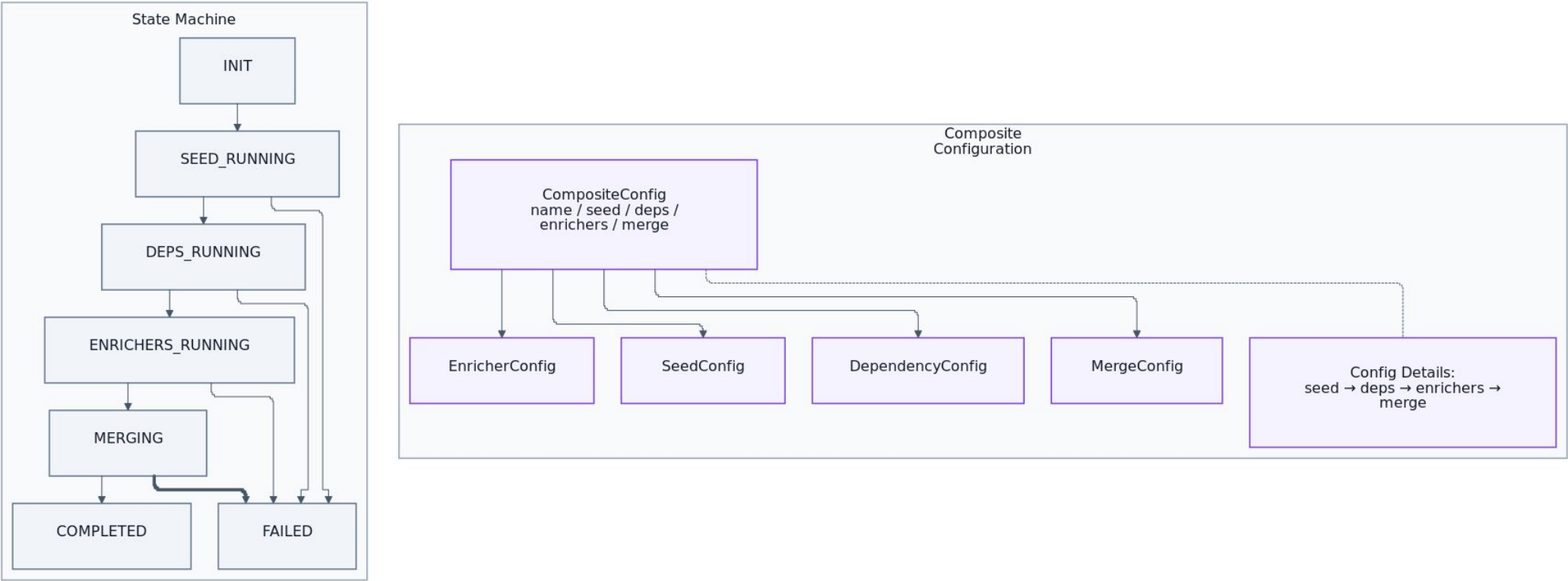
19. Composite Pipeline Architecture

ID: 08-composite-pipeline Type: flowchart Level: System / Component Date: 2026-02-24



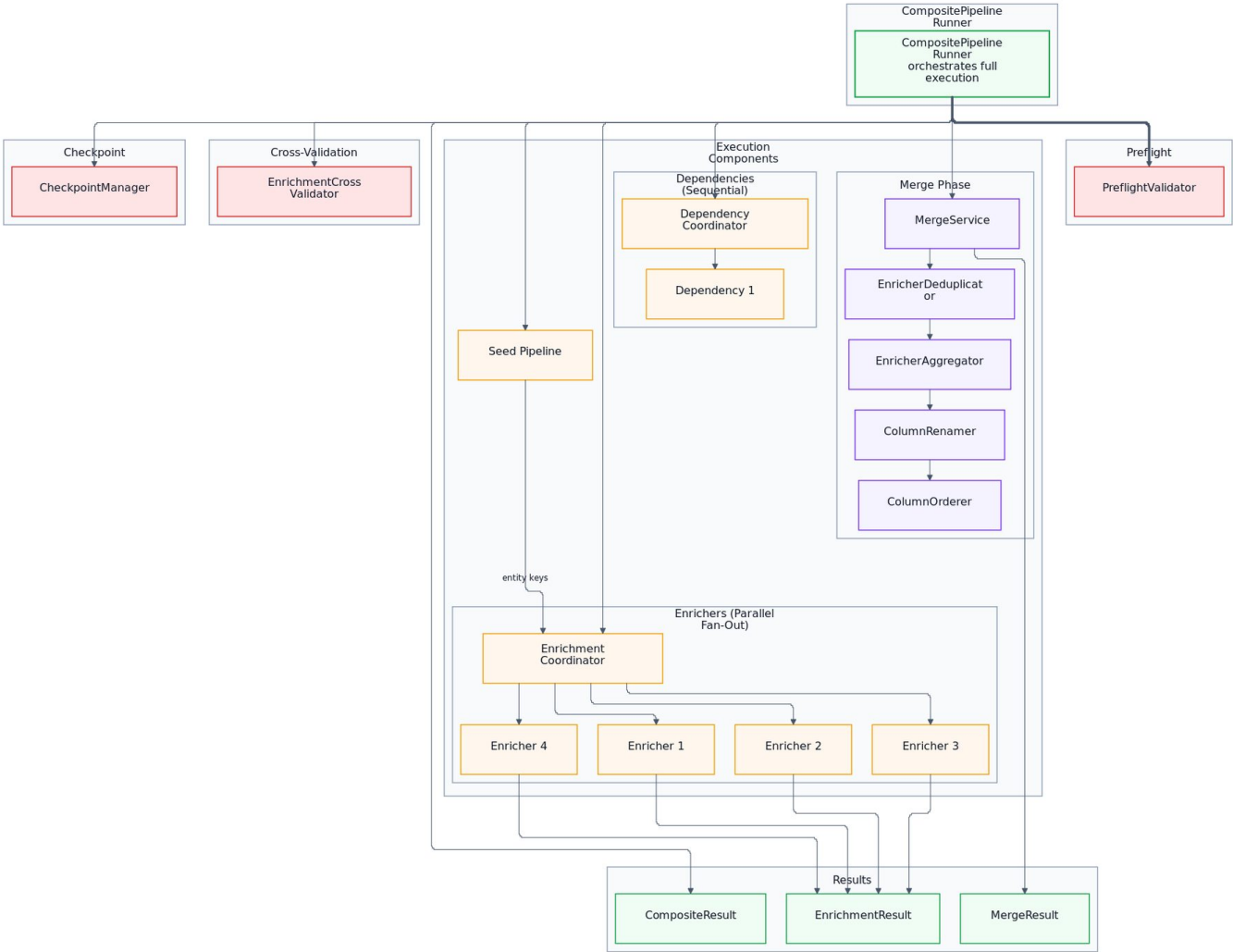
20. Composite Pipeline Configuration & FSM

ID: 08a-composite-config Type: flowchart Level: System / Component Date: 2026-02-27



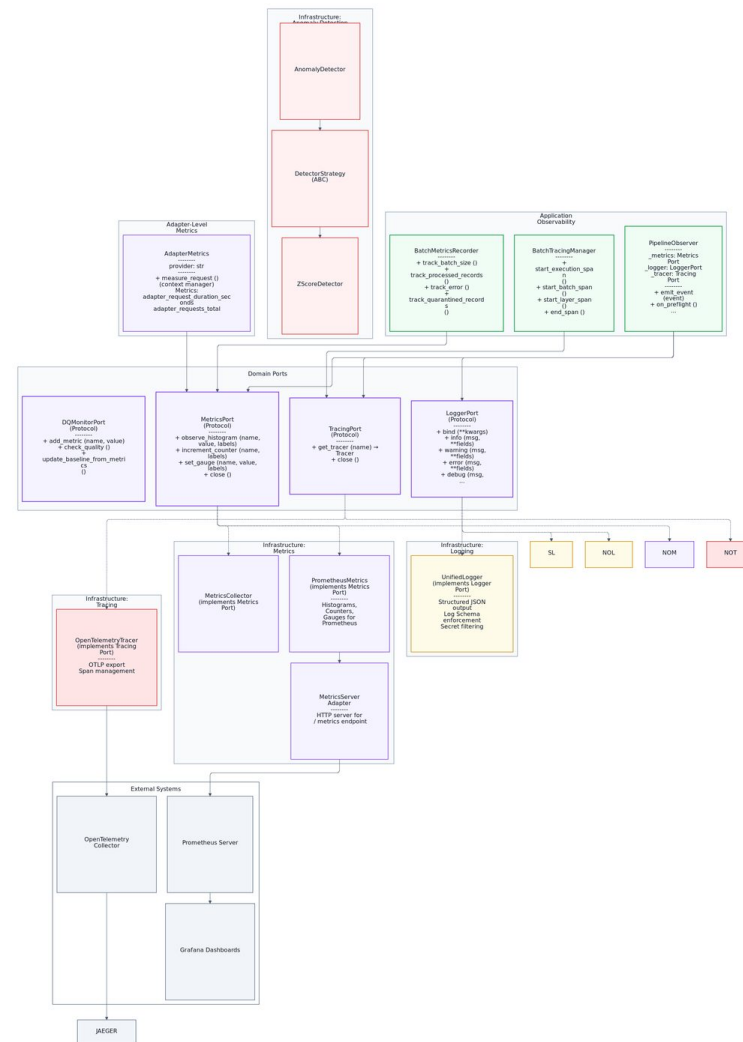
21. Composite Pipeline Execution

ID: 08b-composite-execution Type: flowchart Level: System / Component Date: 2026-02-27



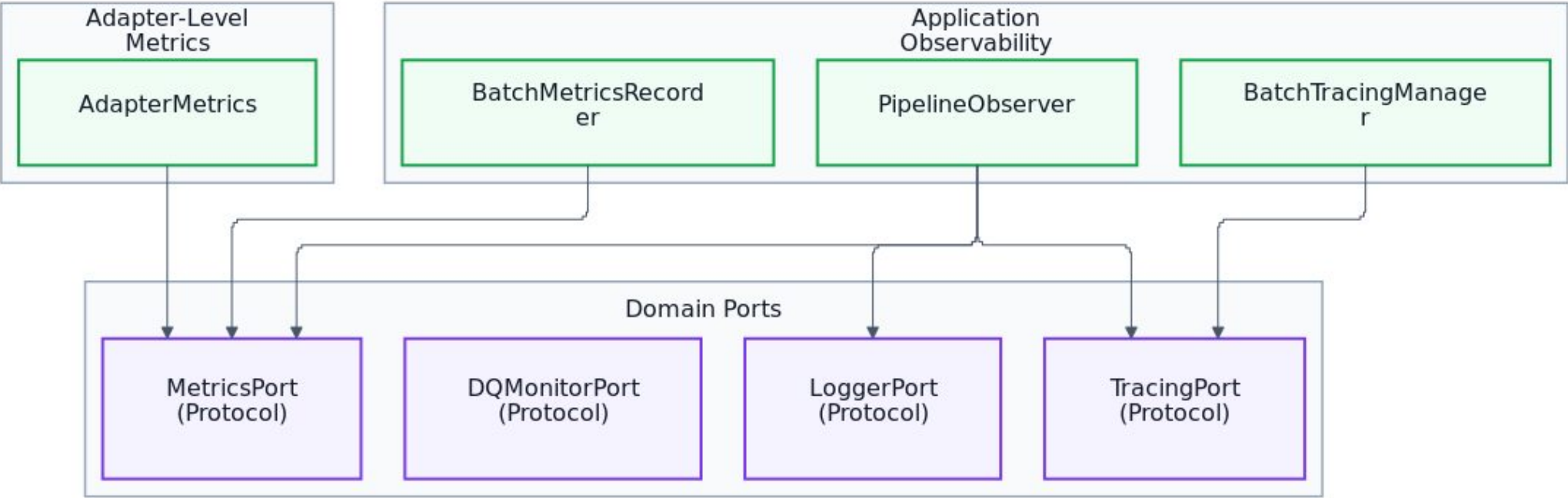
22. Observability Stack

ID: 09-observability-stack Type: flowchart Level: System / Component Date: 2026-02-24



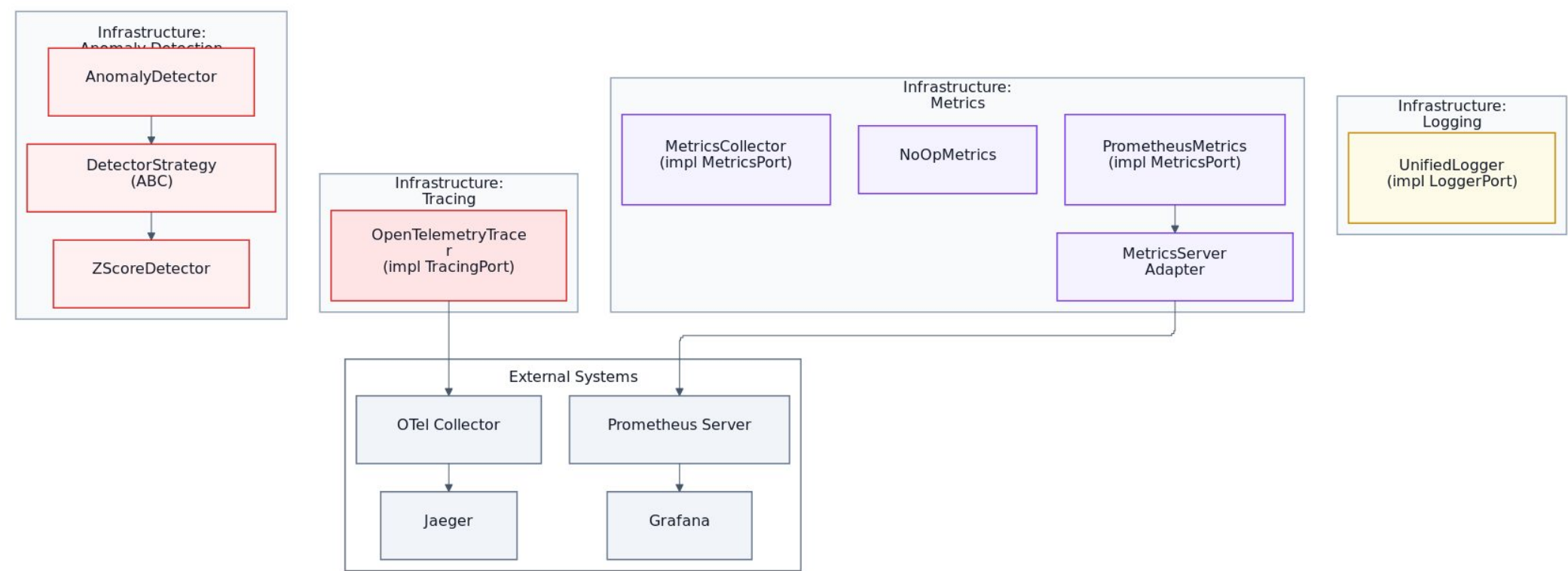
23. Observability: Application Layer

ID: 09a-observability-app Type: flowchart Level: System / Component Date: 2026-02-27



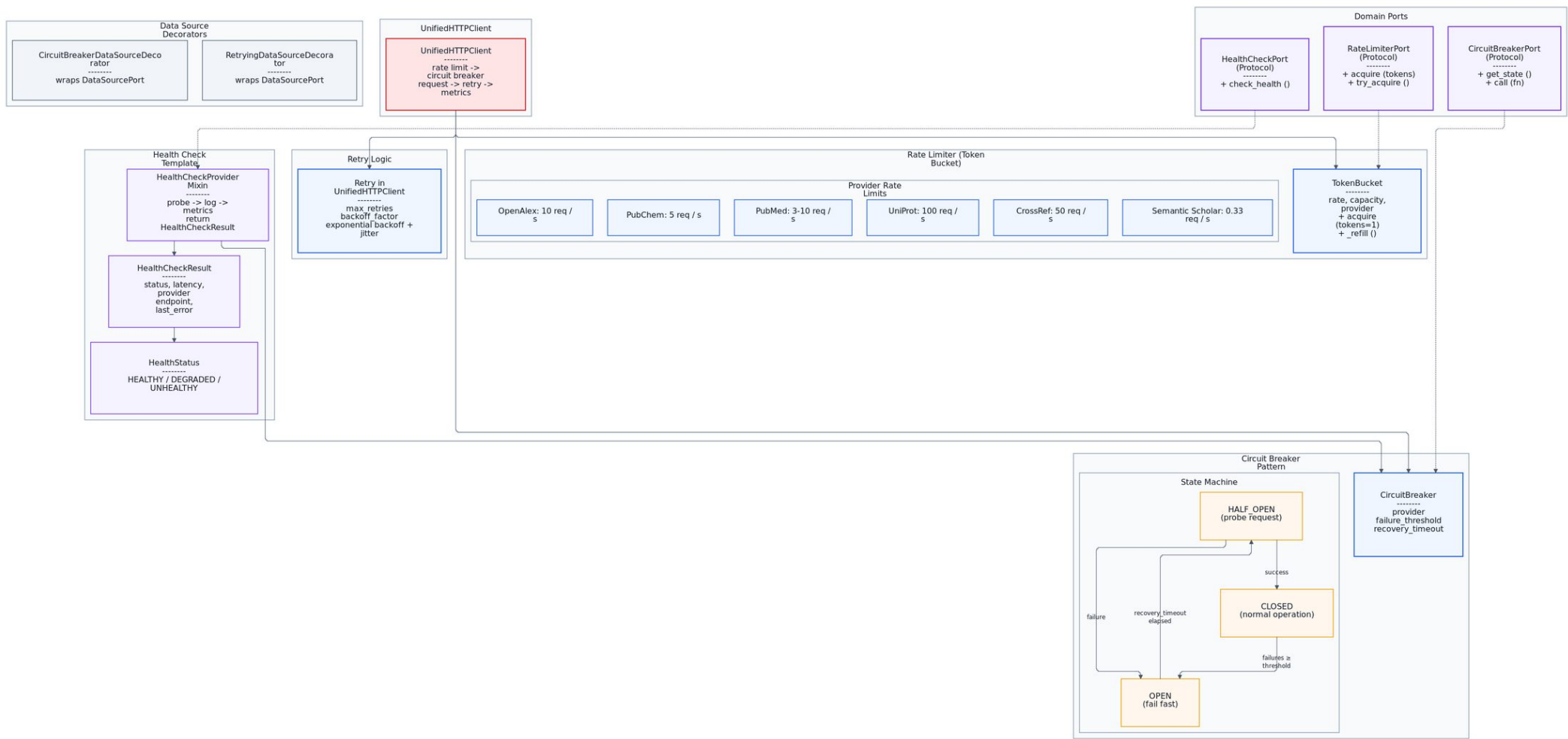
24. Observability: Infrastructure Layer

ID: 09b-observability-infra Type: flowchart Level: System / Component Date: 2026-02-27



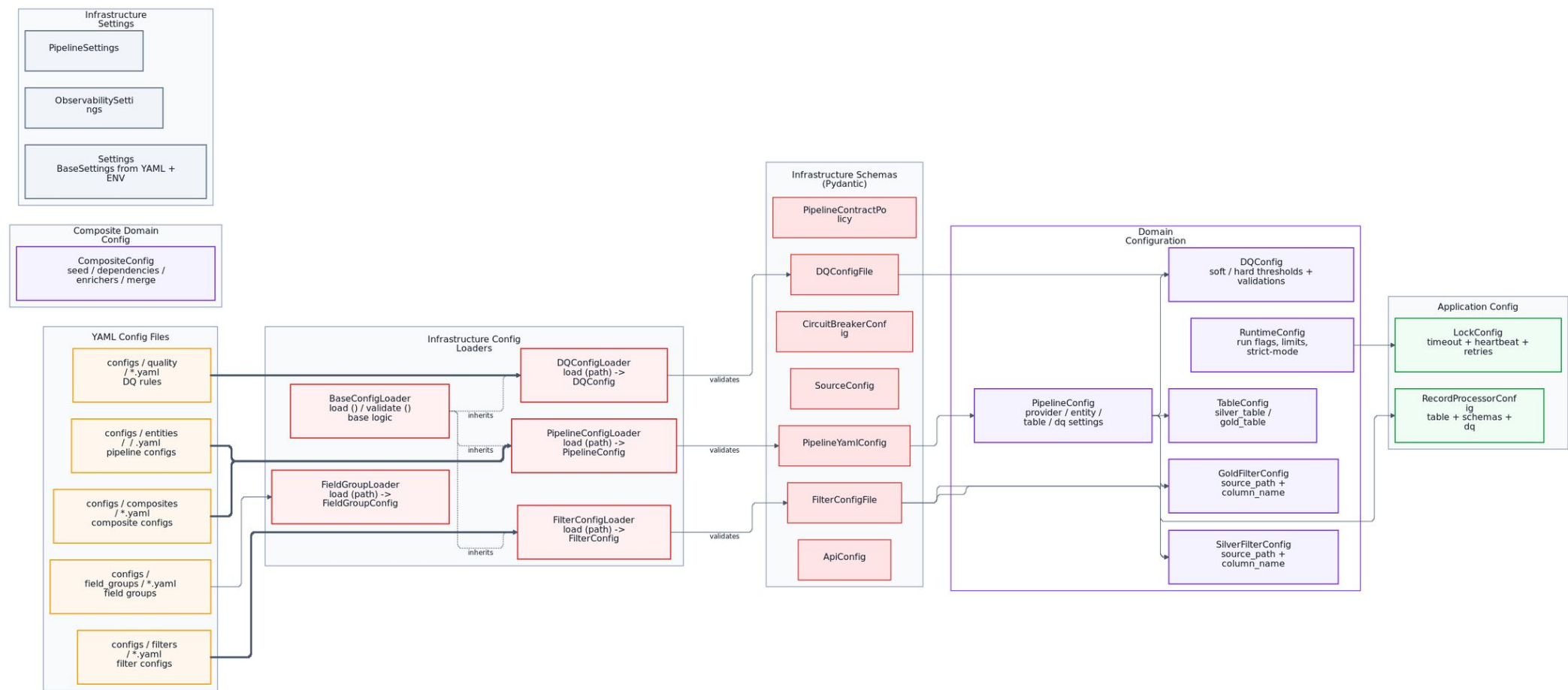
25. Resilience Patterns

ID: 10-resilience-patterns Type: flowchart Level: System / Component Date: 2026-02-24



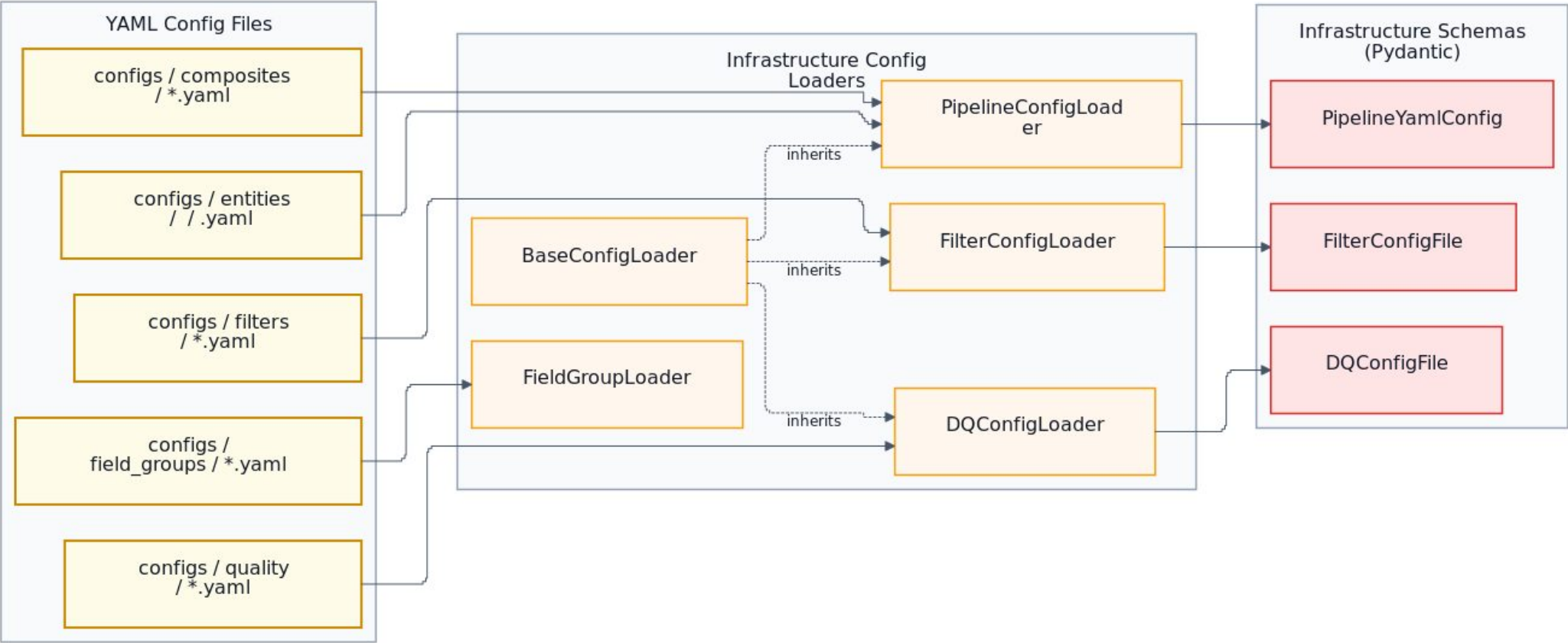
26. Configuration System

ID: 11-configuration-system Type: flowchart Level: System / Component Date: 2026-02-27



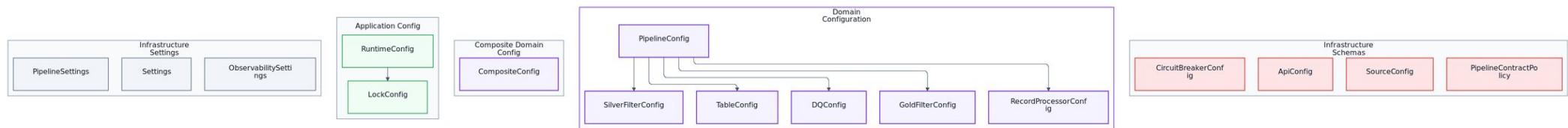
27. Configuration: Loading Pipeline

ID: 11a-config-loading Type: flowchart Level: System / Component Date: 2026-02-27



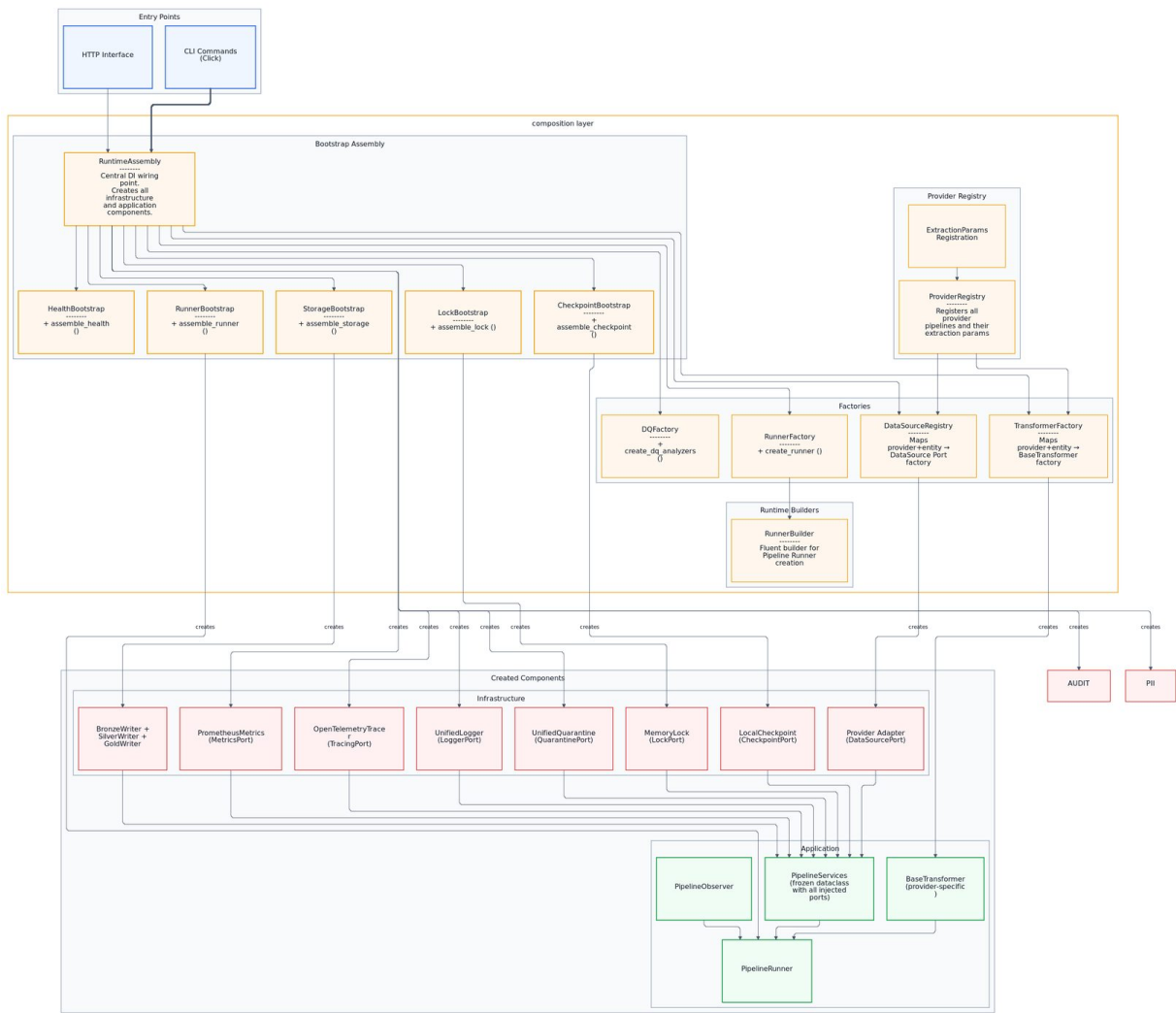
28. Configuration: Domain & Application Config

ID: 11b-config-domain Type: flowchart Level: System / Component Date: 2026-02-27



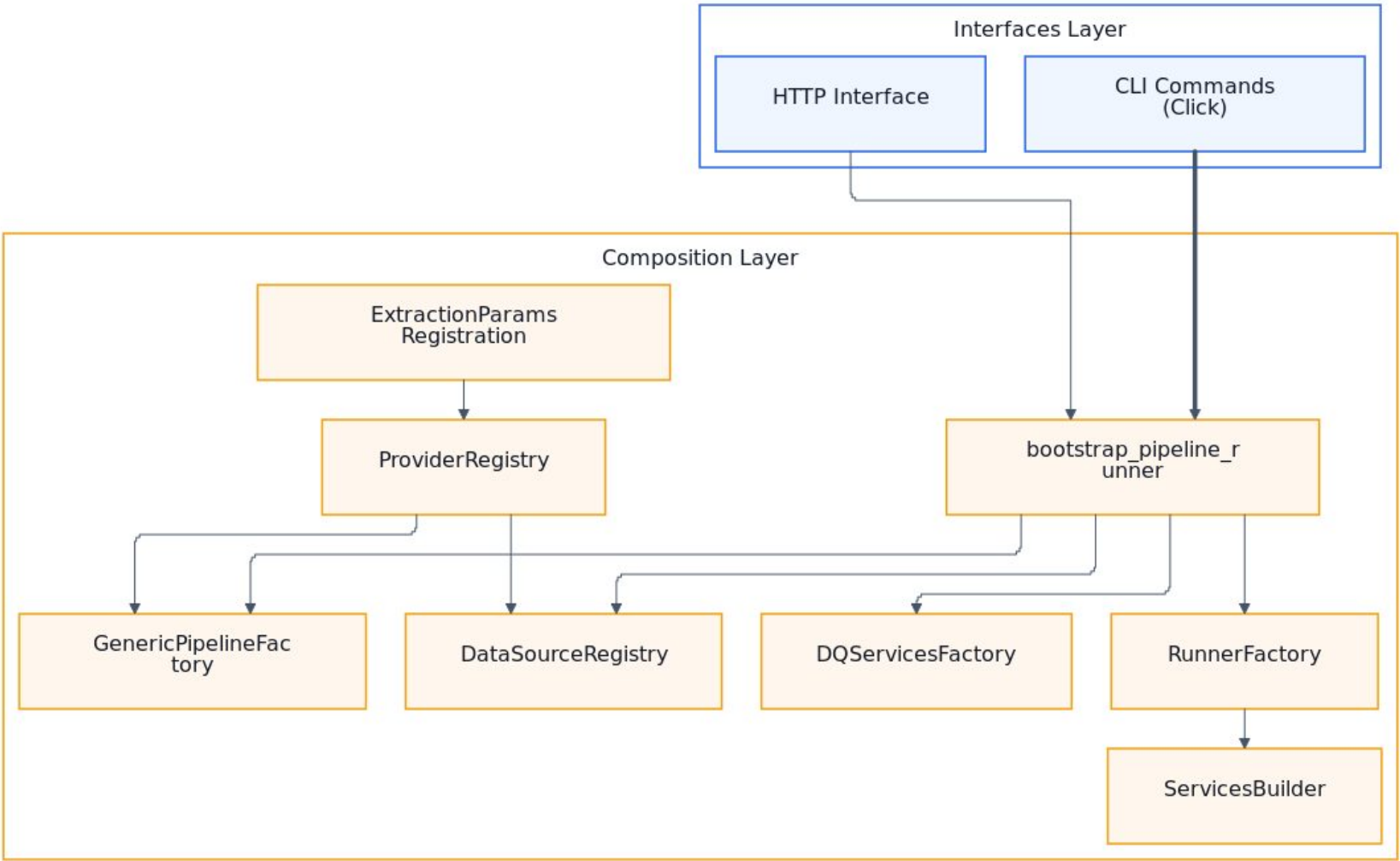
29. Bootstrap / DI Container (Composition Root)

ID: 12-bootstrap-di-container Type: flowchart Level: System / Component Date: 2026-02-24



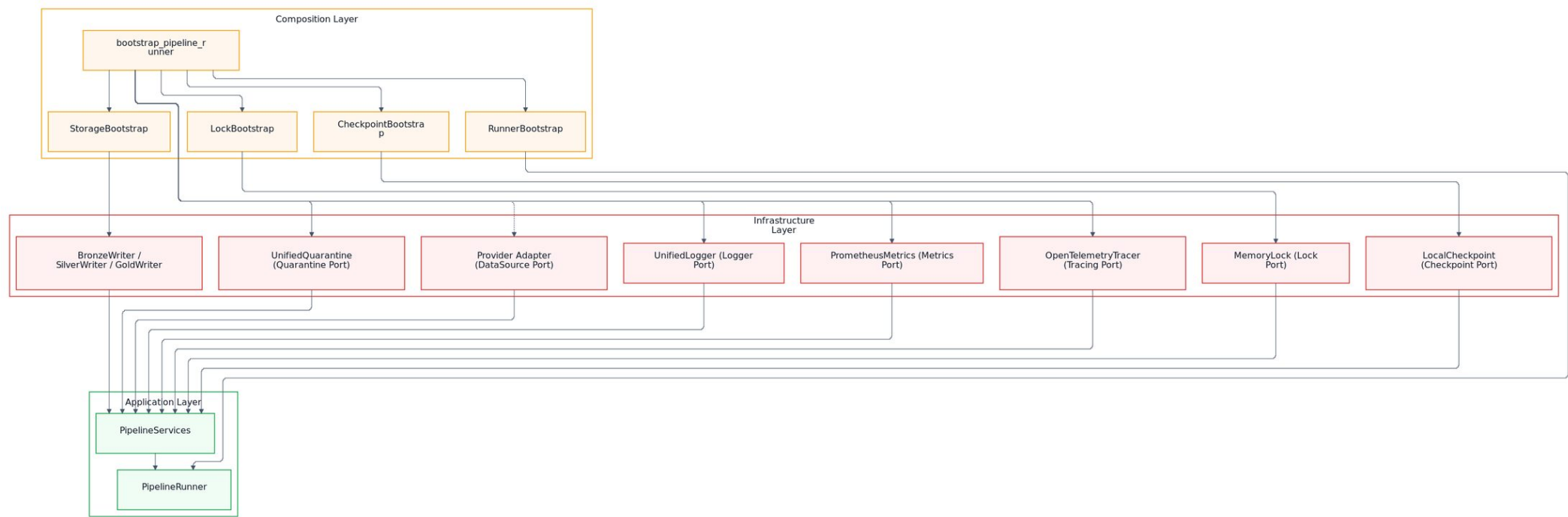
30. Bootstrap: Factories and Registries

ID: 12a-bootstrap-factories Type: flowchart Level: System / Component Date: 2026-02-25



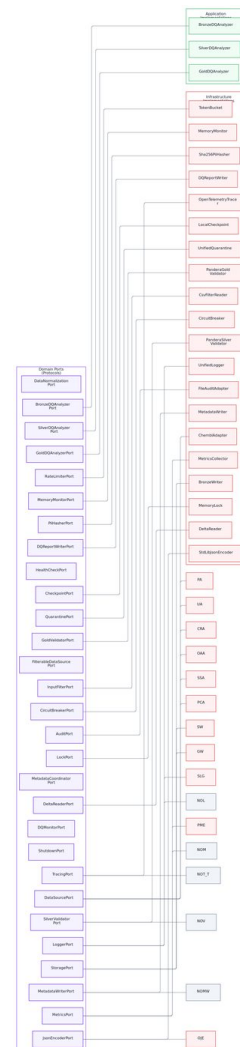
31. Bootstrap: Wiring Graph

ID: 12b-bootstrap-wiring Type: flowchart Level: System / Component Date: 2026-02-25



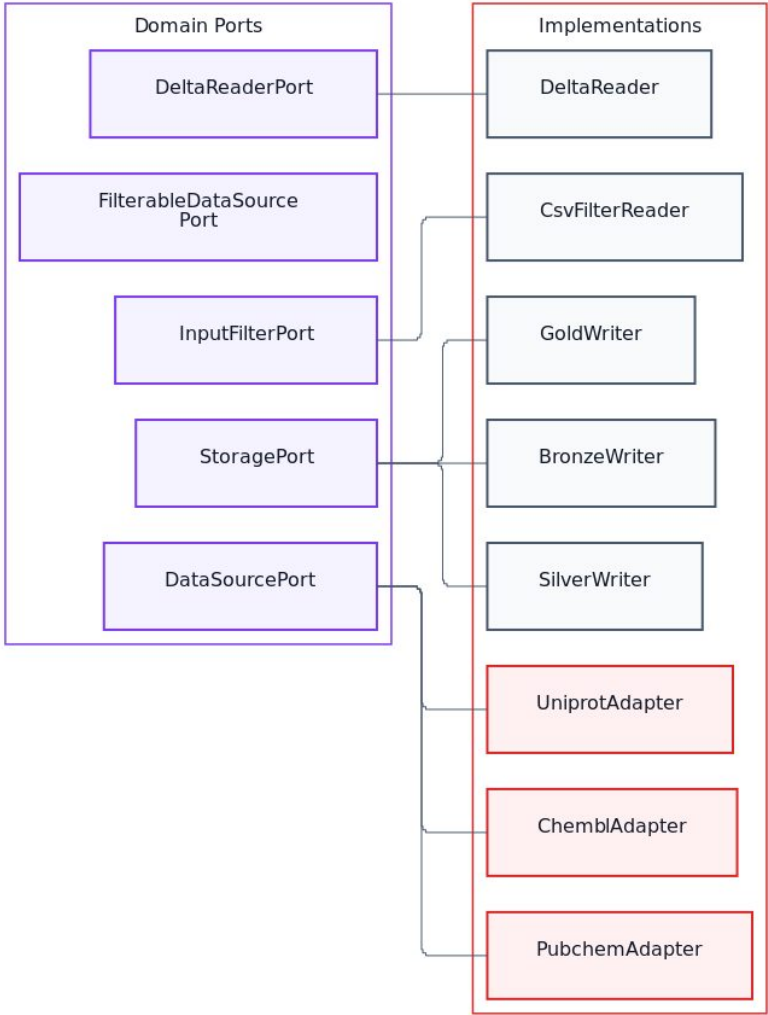
32. Port/Protocol Contracts (Full Map)

ID: 13-port-protocol-contracts Type: flowchart Level: System / Component Date: 2026-02-24



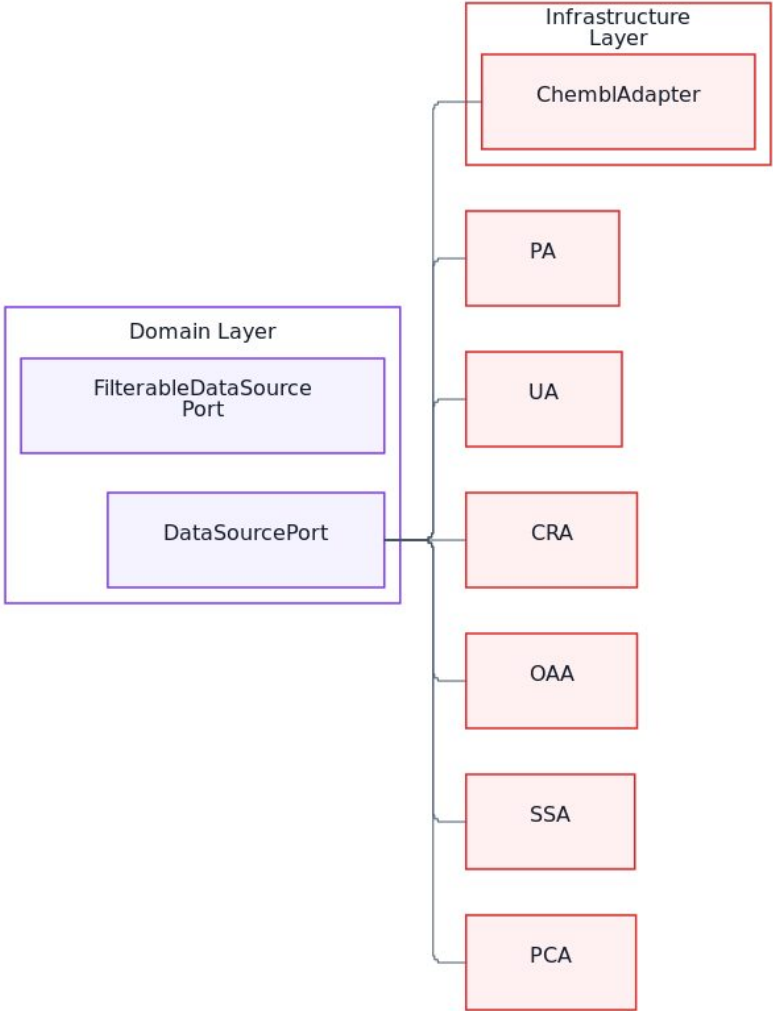
33. DataSource and Storage Ports

ID: 13a-data-storage-ports Type: flowchart Level: System / Component Date: 2026-02-27



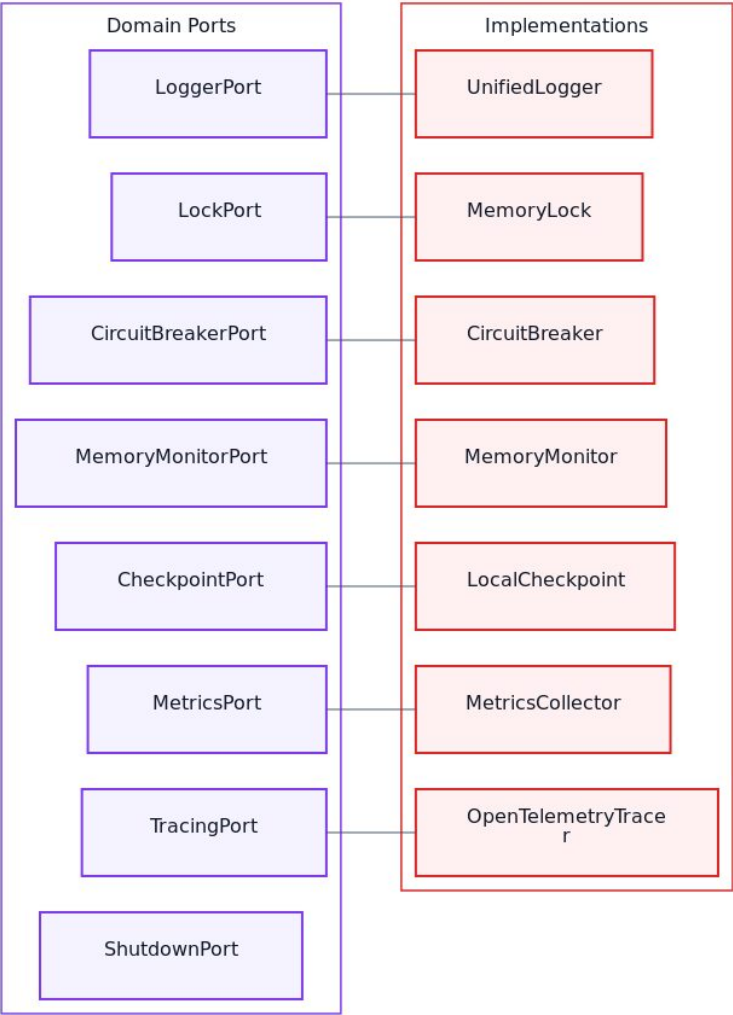
34. Port Contracts: Data Sources

ID: 13a-port-contracts-data-sources Type: flowchart Level: System / Component Date: 2026-02-25



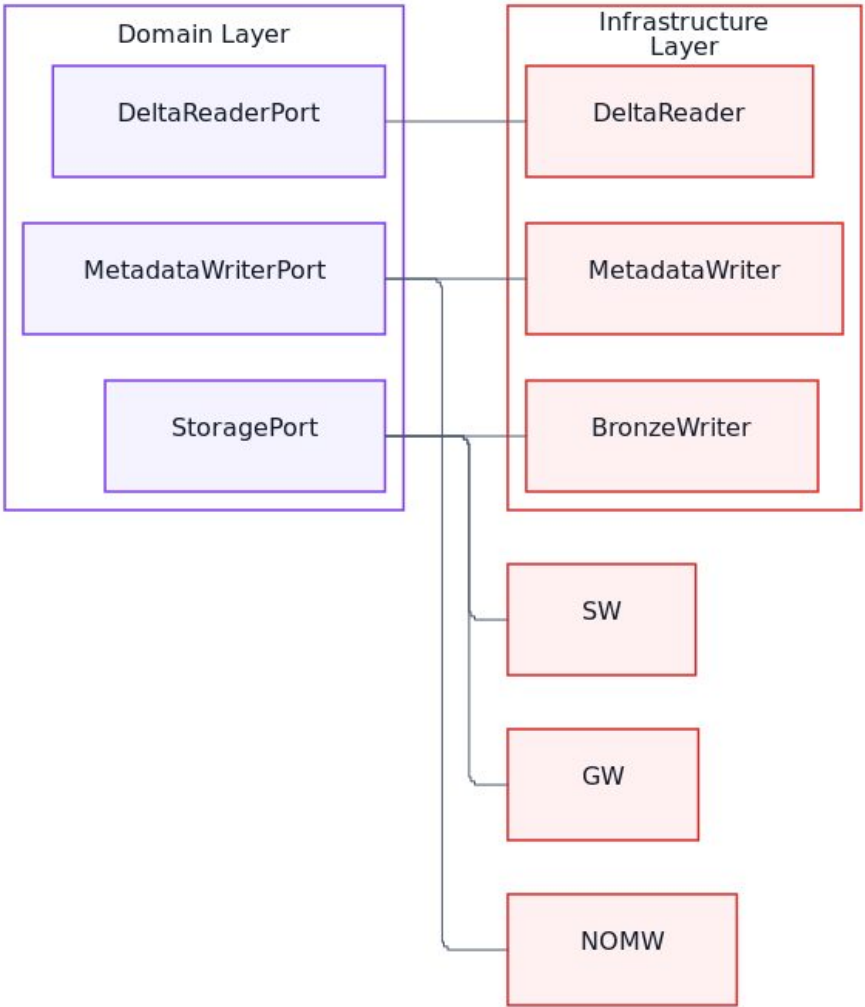
35. Operational and Observability Ports

ID: 13b-operational-ports Type: flowchart Level: System / Component Date: 2026-02-27



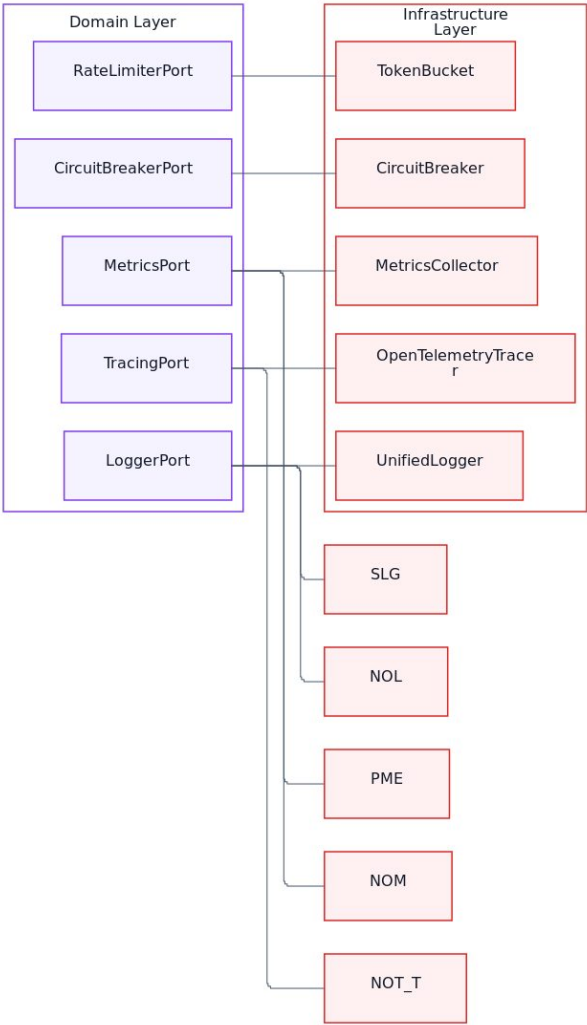
36. Port Contracts: Storage

ID: 13b-port-contracts-storage Type: flowchart Level: System / Component Date: 2026-02-25



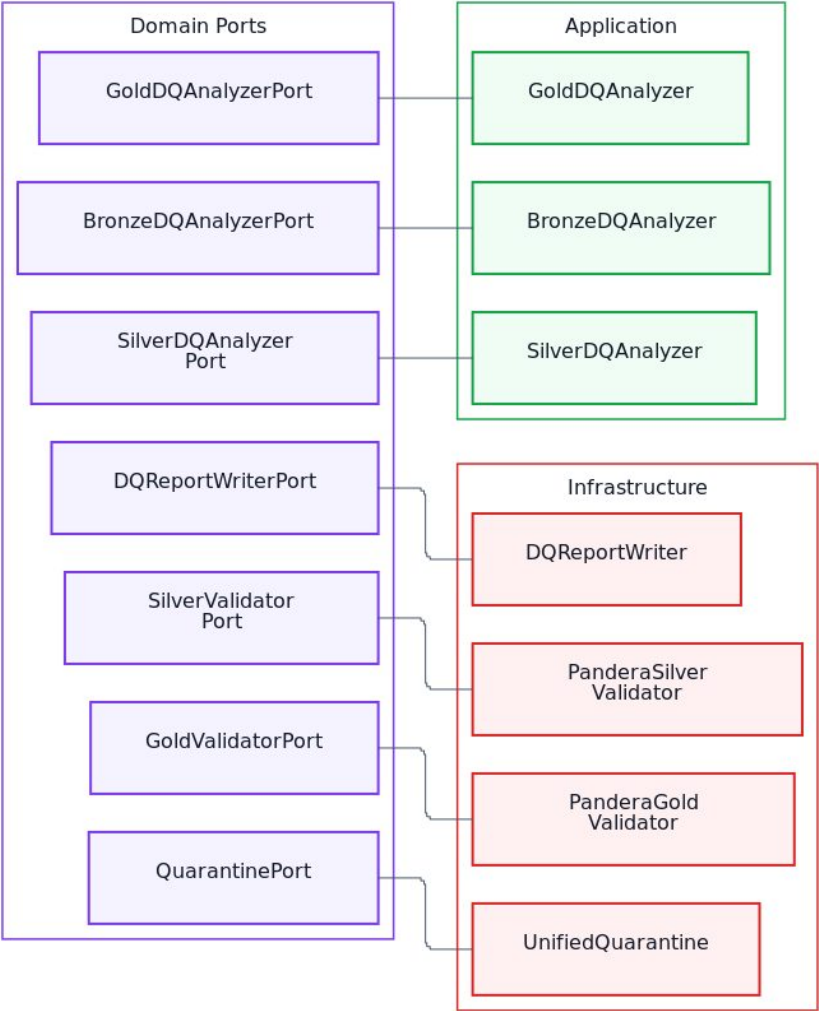
37. Port Contracts: Observability and Resilience

ID: 13c-port-contracts-observability Type: flowchart Level: System / Component Date: 2026-02-25



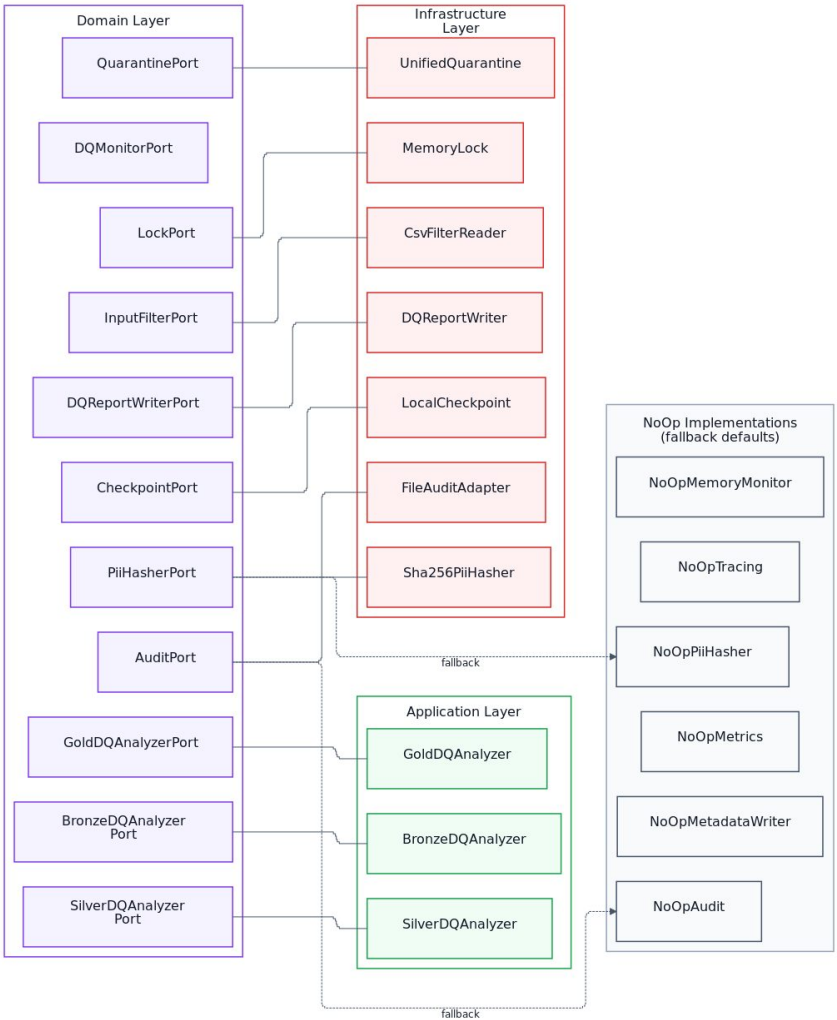
38. Validation and Data Quality Ports

ID: 13c-validation-dq-ports Type: flowchart Level: System / Component Date: 2026-02-27



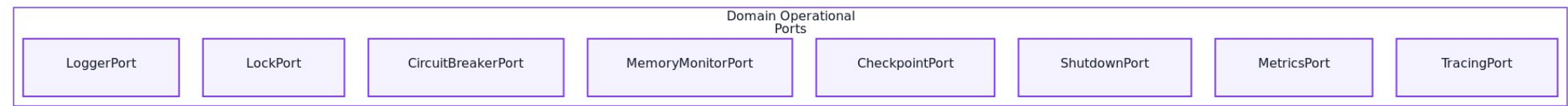
39. Port Contracts: Services and Controls

ID: 13d-port-contracts-services Type: flowchart Level: System / Component Date: 2026-02-25



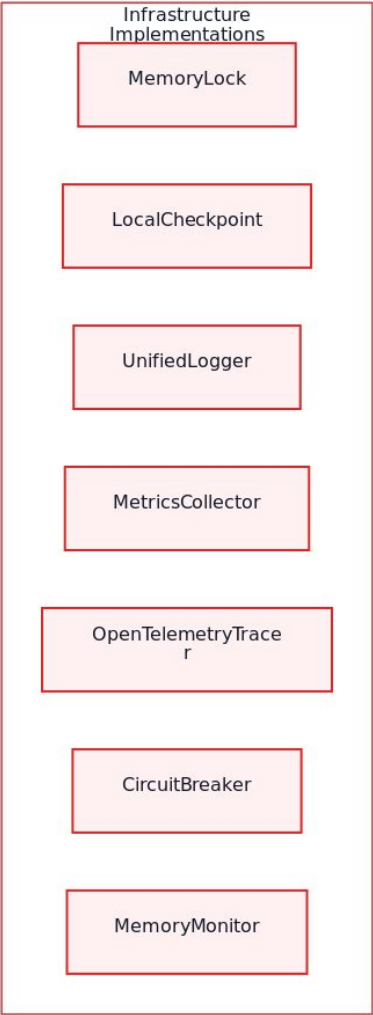
40. Domain Operational Ports

ID: 13e-operational-ports-domain Type: flowchart Level: System / Component Date: 2026-02-27



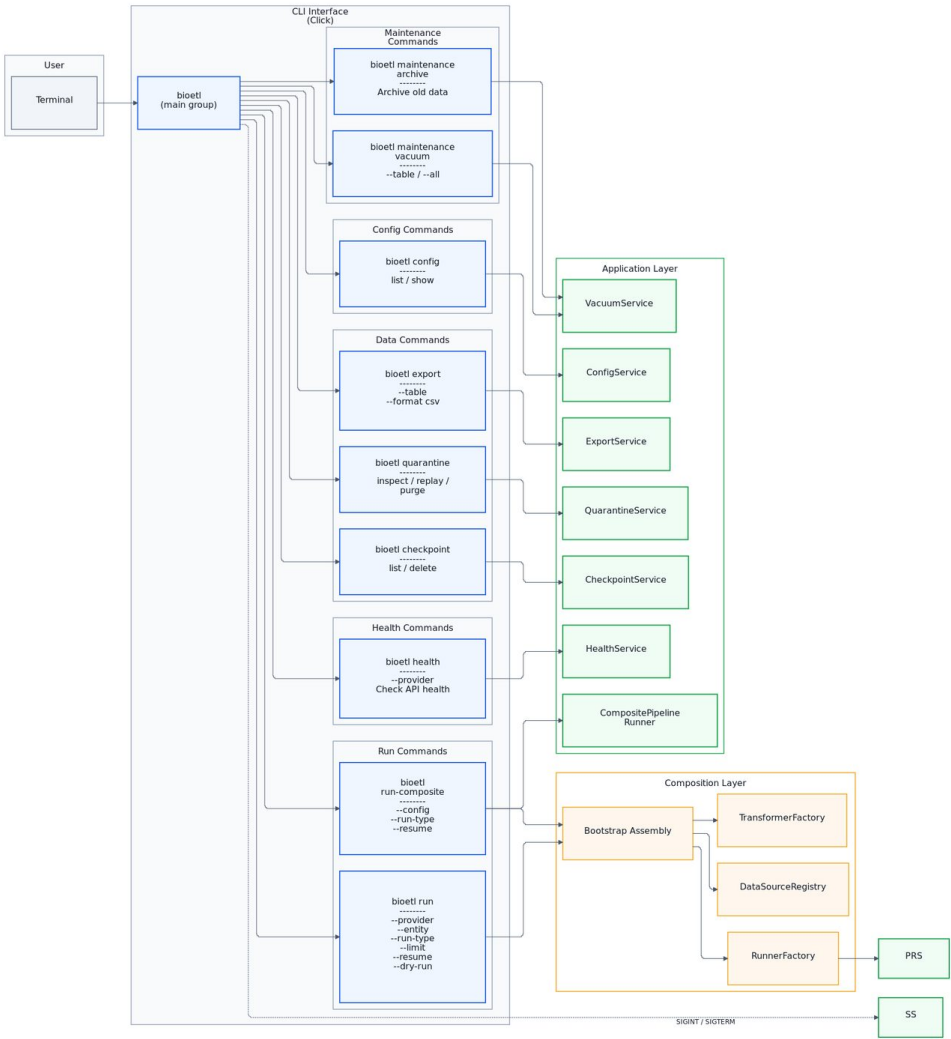
41. Infrastructure Operational Implementations

ID: 13f-operational-ports-infra Type: flowchart Level: System / Component Date: 2026-02-27



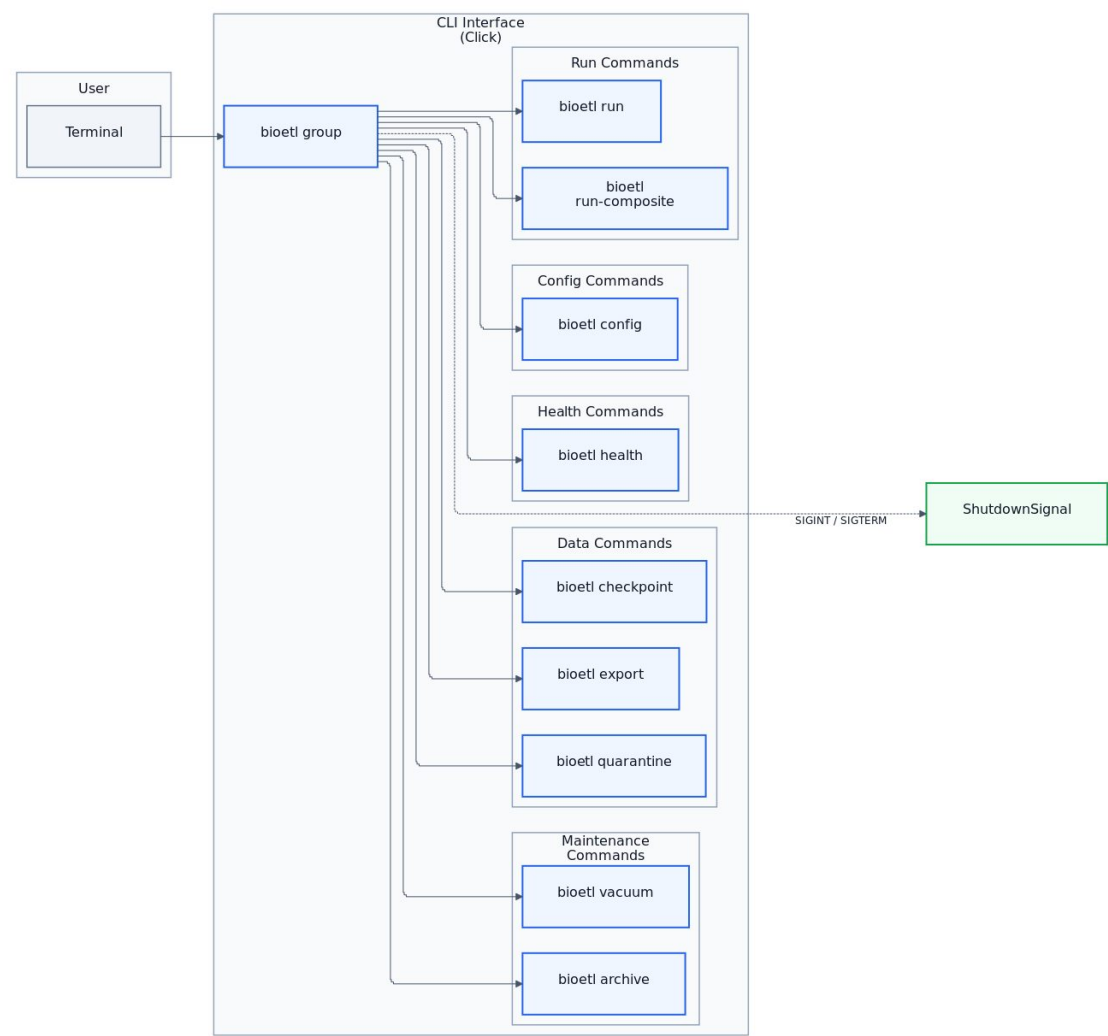
42. CLI / Interface Layer

ID: 14-cli-interface-layer Type: flowchart Level: System / Component Date: 2026-02-24



43. CLI: Command Structure

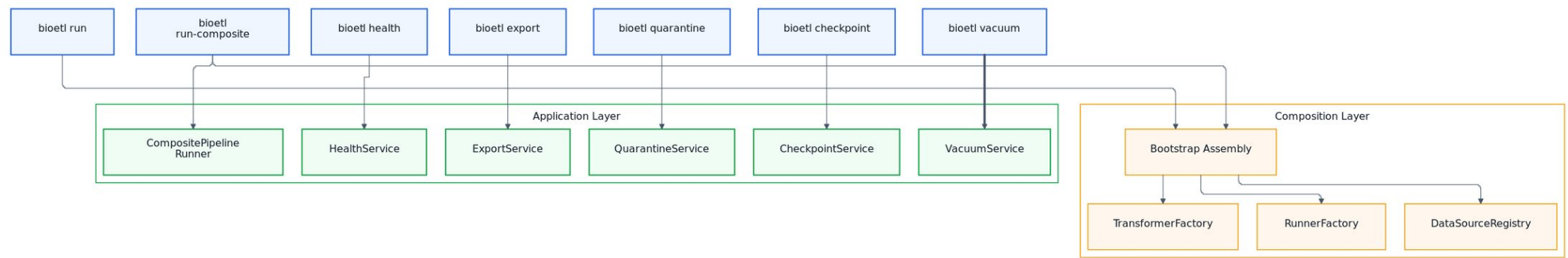
ID: 14a-cli-commands Type: flowchart Level: System / Component Date: 2026-02-27



44. CLI: Routing to Composition & Application

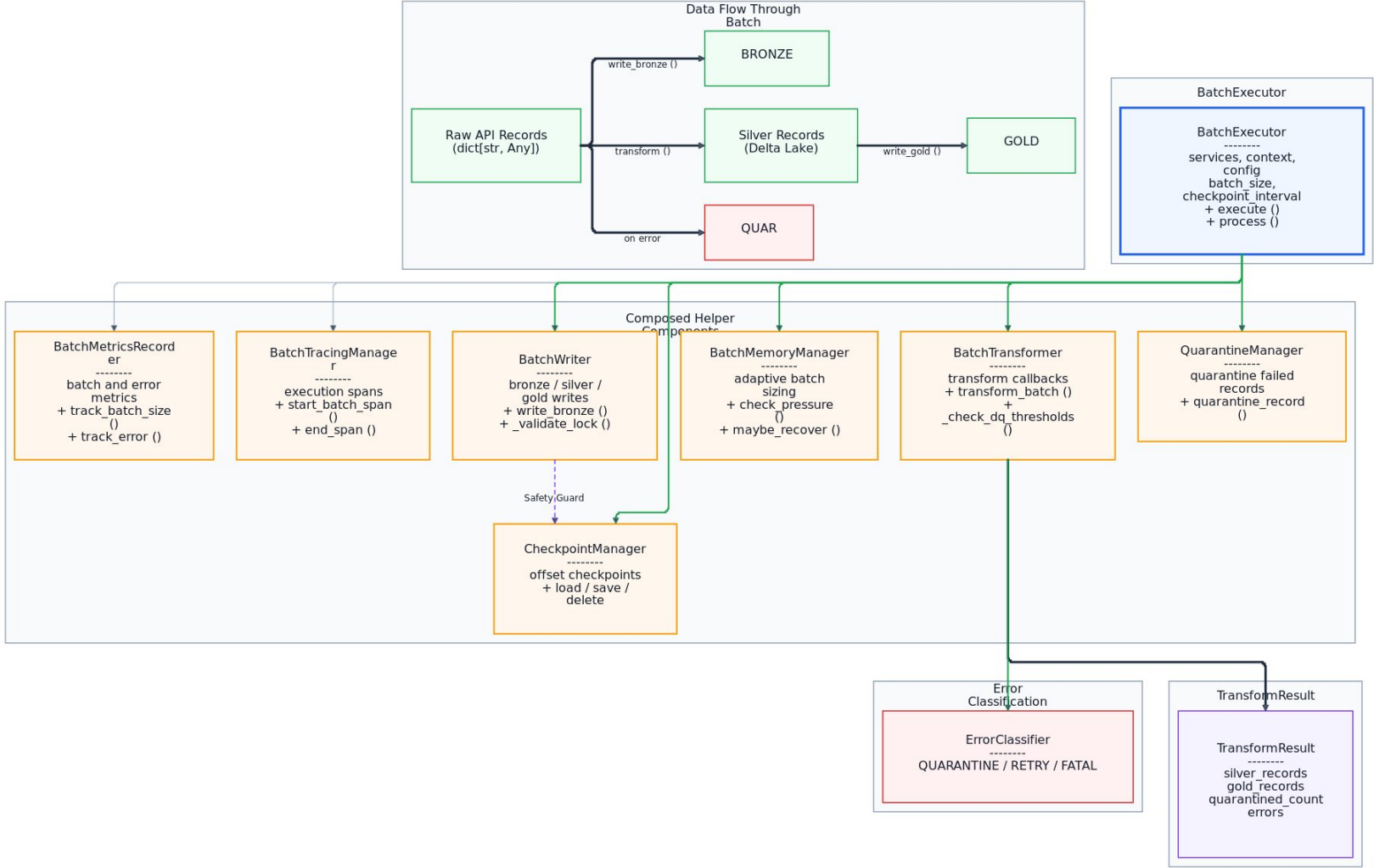
ID: 14b-cli-routing Type: flowchart Level: System / Component Date: 2026-02-27

Source: docs/02-architecture/mmd-diagrams/architecture/14b-cli-routing.mmd | Description: docs/02-architecture/diagram-descriptions/mmd-diagrams/architecture/14b-cli-routing.md



45. BatchExecutor Internal Architecture

ID: 15-batch-executor-internals Type: flowchart Level: System / Component Date: 2026-02-24

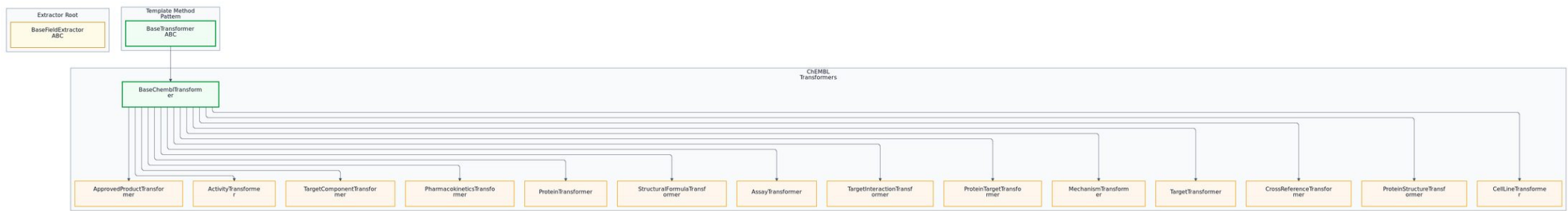


46. Transformer Hierarchy

ID: 16-transformer-hierarchy Type: flowchart Level: System / Component Date: 2026-02-24

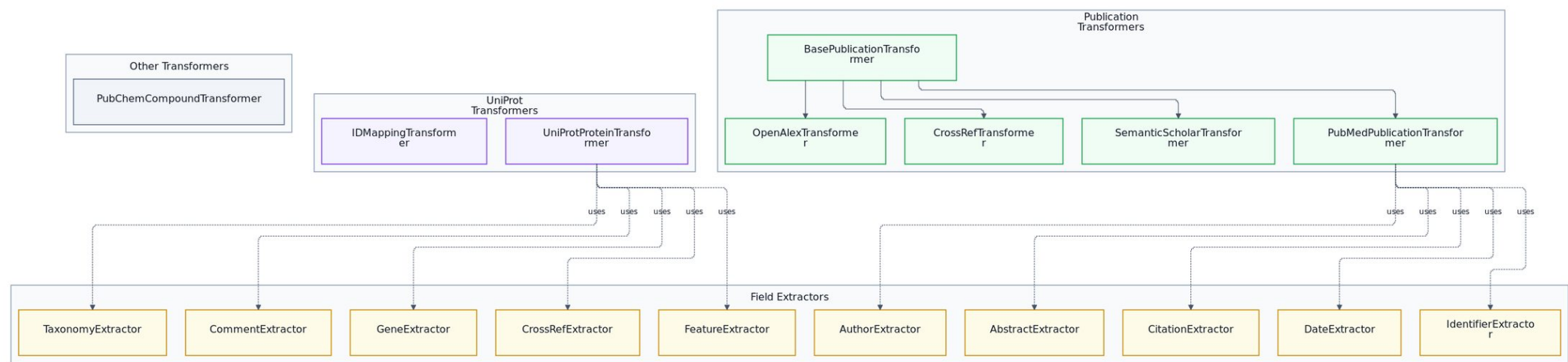
47. Base Transformer and ChEMBL Transformers

ID: 16a-transformer-base Type: flowchart Level: System / Component Date: 2026-02-27



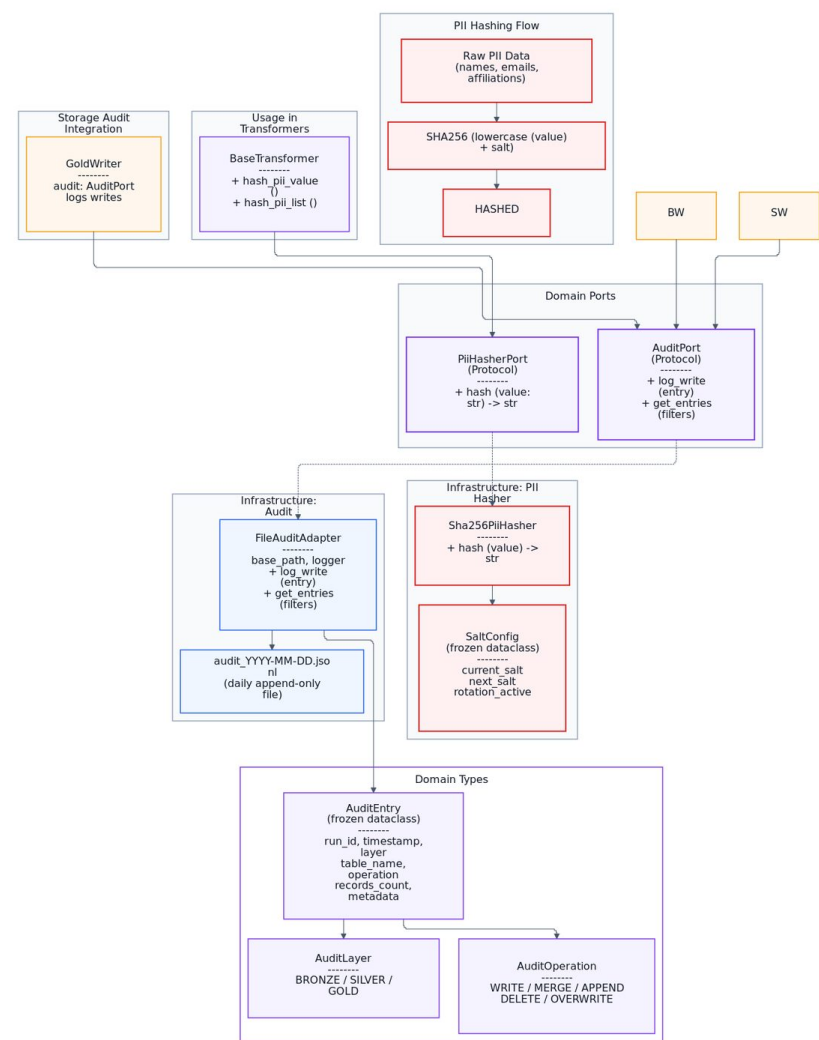
48. Publication, UniProt, Other Transformers and Extractors

ID: 16b-transformer-pub-other Type: flowchart Level: System / Component Date: 2026-02-27



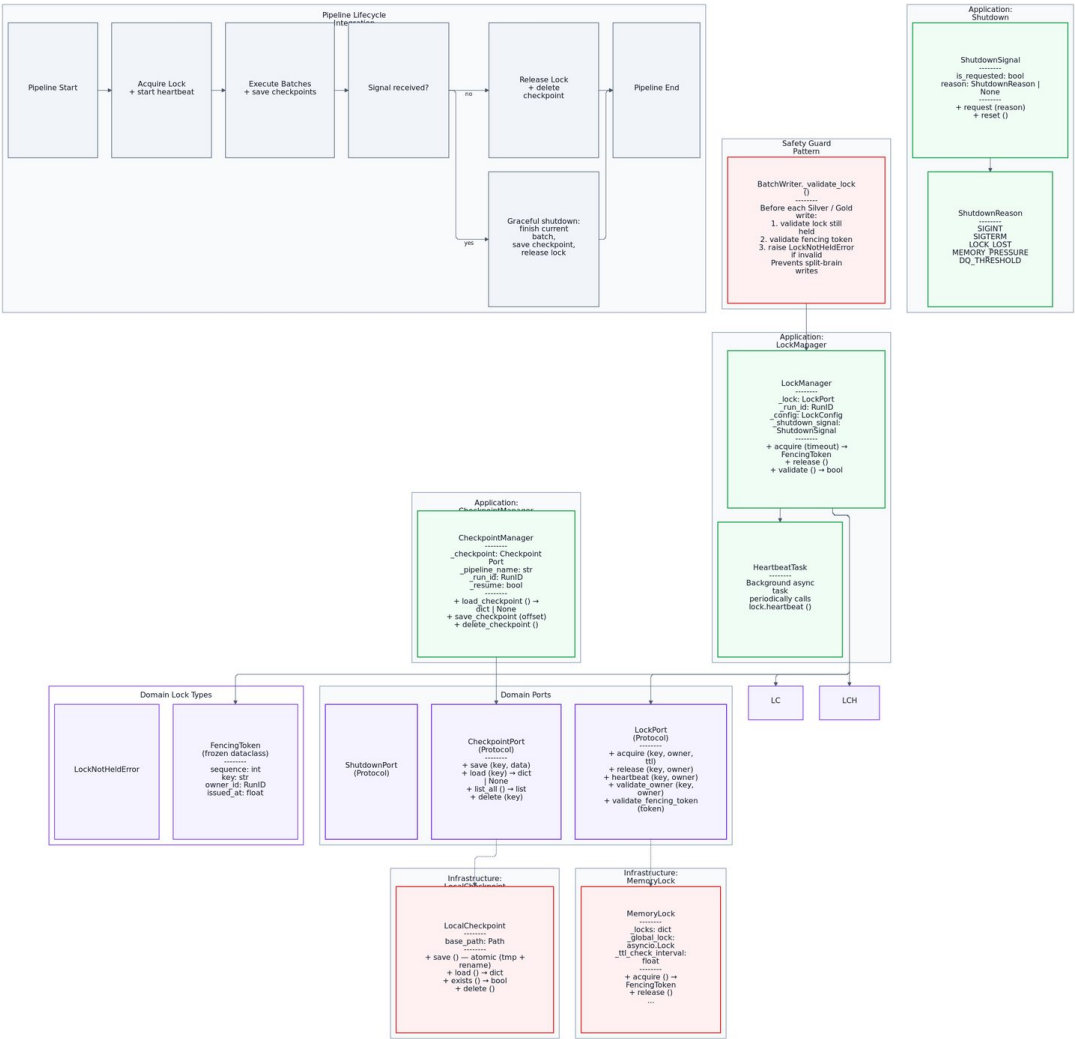
49. Security, PII Hashing, and Audit Trail

ID: 17-security-pii-audit Type: flowchart Level: System / Component Date: 2026-02-24



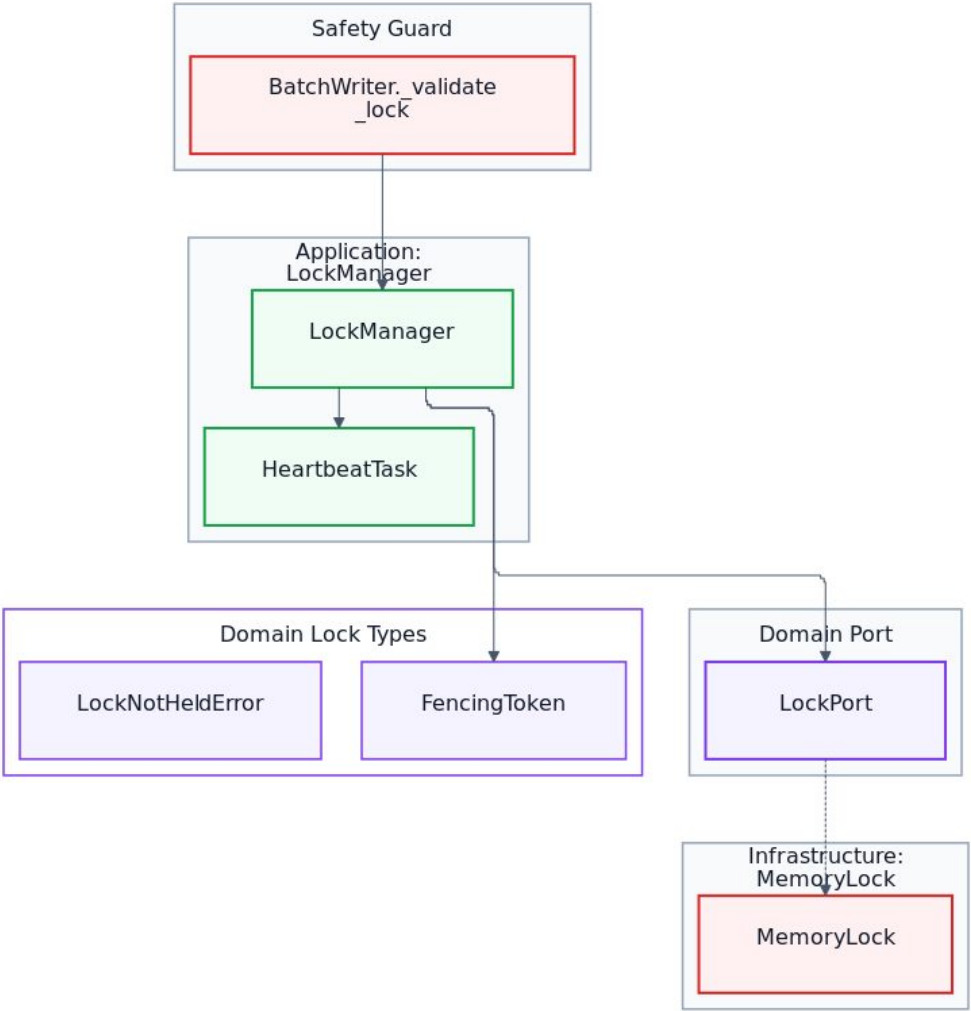
50. Locking, Checkpoint, and Graceful Shutdown

ID: 18-lock-checkpoint-shutdown Type: flowchart Level: System / Component Date: 2026-02-24



51. Lock System

ID: 18a-lock-system Type: flowchart Level: System / Component Date: 2026-02-27



52. Checkpoint and Shutdown System

ID: 18b-checkpoint-shutdown Type: flowchart Level: System / Component Date: 2026-02-27

